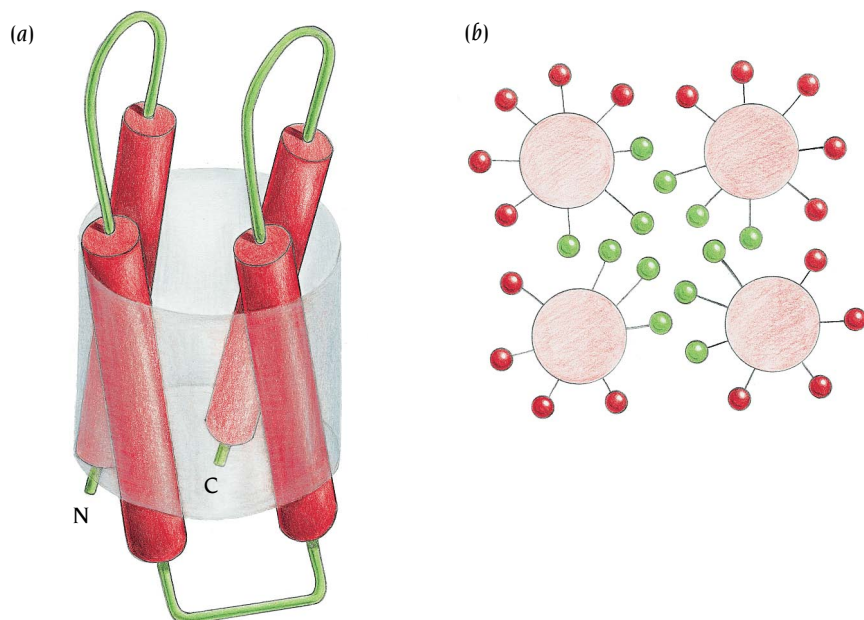
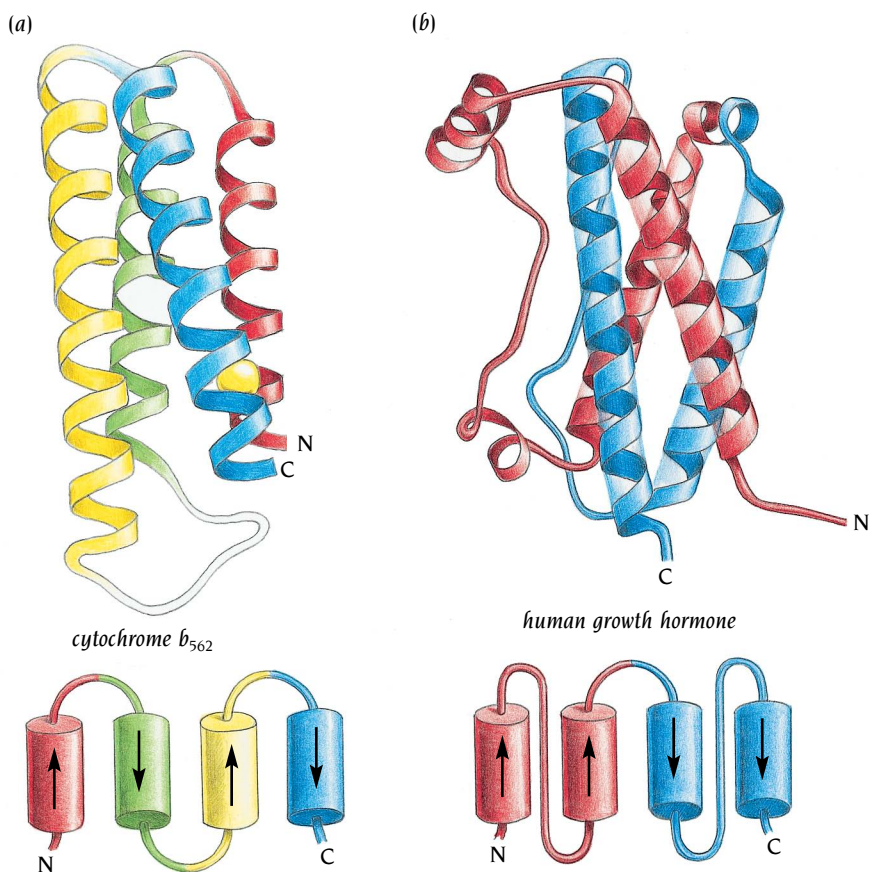




**Figure 3.6** Four-helix bundles frequently occur as domains in  $\alpha$  proteins. The arrangement of the  $\alpha$  helices is such that adjacent helices in the amino acid sequence are also adjacent in the three-dimensional structure. Some side chains from all four helices are buried in the middle of the bundle, where they form a hydrophobic core. (a) Schematic representation of the path of the polypeptide chain in a four-helix-bundle domain. Red cylinders are  $\alpha$  helices. (b) Schematic view of a projection down the bundle axis. Large circles represent the main chain of the  $\alpha$  helices; small circles are side chains. Green circles are the buried hydrophobic side chains; red circles are side chains that are exposed on the surface of the bundle, which are mainly hydrophilic. [(a) Adapted from P.C. Weber and F.R. Salemme, *Nature* 287: 82–84, 1980.]

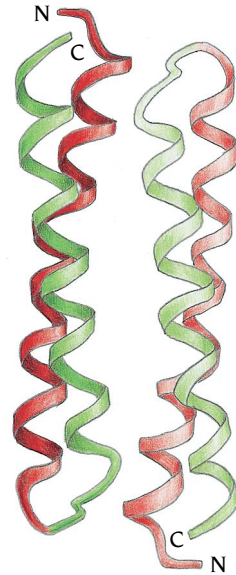
antiparallel fashion. The interaction of this hormone with its receptor is described in Chapter 13.

In most four-helix bundle structures, including those shown in Figure 3.7, the  $\alpha$  helices are packed against each other according to the “ridges in grooves” model discussed later in this chapter. However, there are also examples where coiled-coil dimers packed by the “knobs in holes” model participate in four-helix bundle structures. A particularly simple illustrative example is the Rop protein, a small RNA-binding protein that is encoded by certain plasmids and is involved in plasmid replication. The monomeric subunit of Rop is a polypeptide chain of 63 amino acids built up from two



**Figure 3.7** The polypeptide chains of cytochrome  $b_{562}$  and human growth hormone both form four-helix-bundle structures. In cytochrome  $b_{562}$  (a) adjacent helices are antiparallel, whereas the human growth hormone (b) has two pairs of parallel  $\alpha$  helices joined in an antiparallel fashion.

**Figure 3.8** Schematic diagram of the dimeric Rop molecule. Each subunit comprises two  $\alpha$  helices arranged in a coiled-coil structure with side chains packed into the hydrophobic core according to the “knobs in holes” model. The two subunits are arranged in such a way that a bundle of four  $\alpha$  helices is formed.



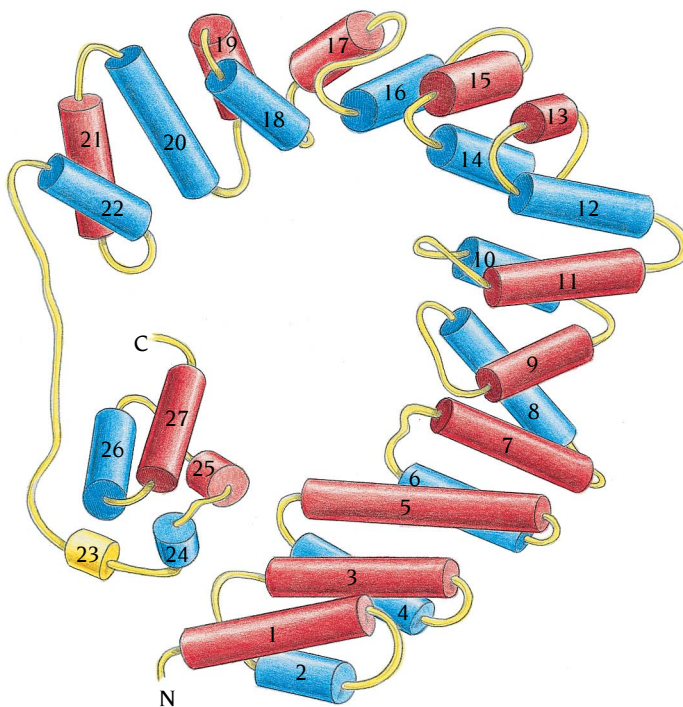
antiparallel  $\alpha$  helices joined by a short loop of three amino acids. The structure of Rop was determined by David Banner at EMBL, Heidelberg, Germany.

The two  $\alpha$  helices of the Rop subunit are arranged as an antiparallel coiled-coil in which the hydrophobic side chains are packed against each other according to the “knobs in holes” model. Two such subunits, each with the same structure, form the dimeric Rop molecule in which the subunits are arranged as a bundle of four  $\alpha$  helices with their long axes aligned (Figure 3.8). The two dimers pack against each other according to the “ridges in grooves” model. The helix-loop-helix (HLH) family of transcription factors, discussed in Chapter 10, is another example of a four-helix bundle structure involving coiled-coil helices.

### *Alpha-helical domains are sometimes large and complex*

The structures of several enzymes are known in which a long polypeptide chain of 300–400 amino acids is arranged in more than 20  $\alpha$  helices packed together in a complex pattern to form a globular domain. One such enzyme is a bacterial muramidase that is involved in the metabolism of peptidoglycans, which form part of the bacterial cell wall. The structure of this enzyme was determined by Bauke Dijkstra and colleagues in Groningen, Netherlands, as a basis for the design of specific inhibitors to the enzyme, which might lead eventually to novel types of antibacterial drugs.

The polypeptide chain of this monomeric enzyme has 618 amino acids, of which the N-terminal 450 residues form one  $\alpha$ -helical domain. This domain is built up from 27  $\alpha$  helices arranged in a two-layered ring with a right-handed superhelical twist (Figure 3.9). The ring has a large central hole, like in a doughnut, with a diameter of about 30 Å. The remaining residues form the catalytic domain that lies on top of the ring. The function of the



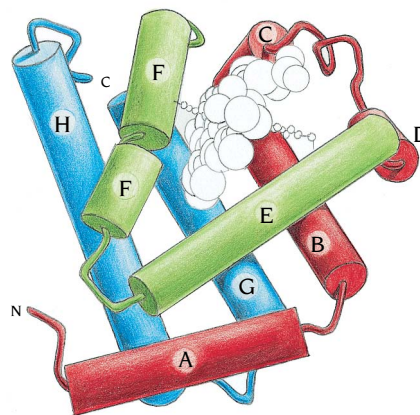
**Figure 3.9** Schematic diagram of the structure of one domain of a bacterial muramidase, comprising 450 amino acid residues. The structure is built up from 27  $\alpha$  helices arranged in a two-layered ring. The ring has a large central hole, like a doughnut, with a diameter of about 30 Å.

doughnut-shaped domain is not known, but its shape may be required for the specificity of the catalytic reaction *in vivo*.

### *The globin fold is present in myoglobin and hemoglobin*

One of the most important  $\alpha$  structures is the **globin fold**. This fold has been found in a large group of related proteins, including myoglobin, hemoglobins, and the light-capturing assemblies in algae, the phycocyanins. The functional and evolutionary aspects of these structures will not be discussed in this book; instead, we will examine some features that are of general structural interest.

The pairwise arrangements of the sequential  $\alpha$  helices in the globin fold are quite different from the antiparallel organization found in the four-helix-bundle  $\alpha$  structures. The globin structure is a bundle of eight  $\alpha$  helices, usually labeled A–H, connected by rather short loop regions and arranged so that the helices form a pocket for the active site, which in myoglobin and the hemoglobins binds a heme group (Figure 3.10). The lengths of the  $\alpha$  helices vary considerably, from 7 residues in the shortest helix (C) to 28 in the longest helix (H) in myoglobin. In the globin fold the  $\alpha$  helices wrap around the core in different directions so that sequentially adjacent  $\alpha$  helices are usually not adjacent to each other in the structure. The only exceptions are the last two  $\alpha$  helices (G and H), which form an antiparallel pair with extensive packing interactions between them. All other packing interactions are formed between pairs of  $\alpha$  helices that are not sequentially adjacent. Because the globin fold is not built up from an assembly of smaller motifs, it is quite difficult to visualize conceptually in spite of its relatively small size and simplicity.



**Figure 3.10** Schematic diagram of the globin domain. The eight  $\alpha$  helices are labeled A–H. A–D are red, E and F green, and G and H blue. The heme group is shown in white. (Adapted from originals provided by A. Lesk.)

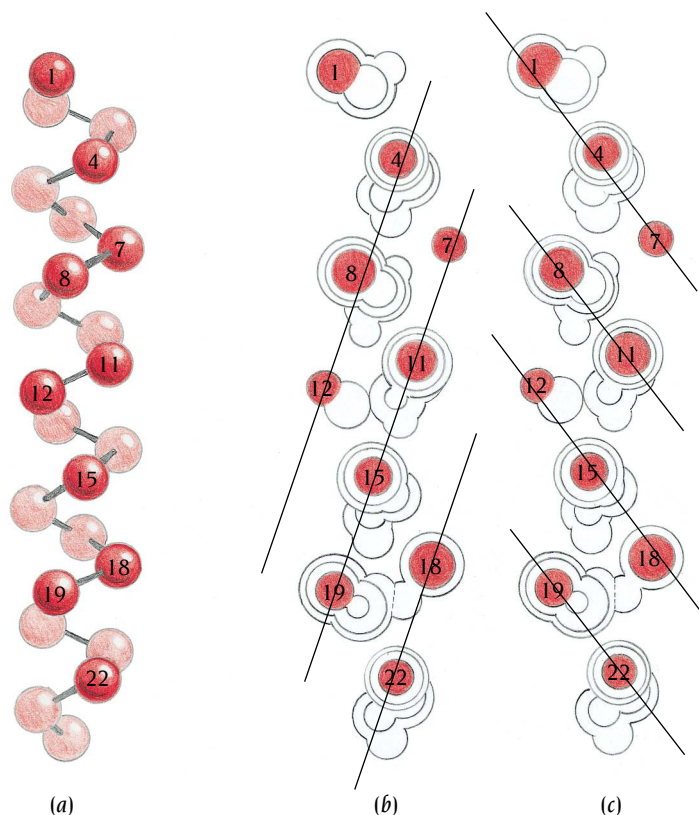
### *Geometric considerations determine $\alpha$ -helix packing*

When we compare the arrangements of the  $\alpha$  helices in coiled-coil structures (see Figure 3.1), in the four-helix-bundle structure (see Figure 3.8), and in the globin fold (see Figure 3.10), it is obvious that the geometry of  $\alpha$ -helix packing is quite different. We described earlier the way that the side chains of coiled-coil  $\alpha$  helices pack according to the “knobs in holes” model. In contrast, other  $\alpha$ -helical structures pack their  $\alpha$  helices according to a “ridges in grooves” model. In the four-helix bundle the  $\alpha$  helices pack almost parallel, or antiparallel, to each other, with an angle of about  $20^\circ$  between the helical axes. In the globin fold the angles between the helical axes are usually larger, in most cases around  $50^\circ$ . These are the two main ways that  $\alpha$  helices pack against each other in the “ridges in grooves” model, a packing motif dictated by the geometry of the surfaces of  $\alpha$  helices.

### *Ridges of one $\alpha$ helix fit into grooves of an adjacent helix*

Since the side chains of an  $\alpha$  helix are arranged in a helical row along the surface of the helix, they form ridges separated by shallow furrows, or grooves, on the surface. Alpha helices pack with the ridges on one helix packing into the grooves of the other and vice versa. The ridges and grooves are formed by amino acids that are usually three or four residues apart. This is illustrated in Figure 3.11, which shows slices through the surface of a polyalanine  $\alpha$  helix on which the directions of the ridges are marked. In contrast to the ridges and grooves of the DNA double helix described in Chapter 7, which are formed by the sugar-phosphate main-chain atoms, those of an  $\alpha$  helix are formed by the amino acid side chains. The detailed geometry of the ridges and grooves of an  $\alpha$  helix is thus dependent not only on the geometry of the helix but also on the actual amino acid sequence.

The most common way of packing  $\alpha$  helices is by fitting the ridges formed by a row of residues separated in sequence by four in one helix into the same type of grooves in the other helix. In this case the ridges and



**Figure 3.11** The side chains on the surface of an  $\alpha$  helix form ridges separated by grooves, as schematically illustrated here. (a) An  $\alpha$  helix with each residue represented by the first atom in the side chain,  $C_{\beta}$ . (b) The surface relief of a polyaniline  $\alpha$  helix in the orientation shown in (a). Sections are cut through a space-filling model and superimposed. The residue numbers are placed on the side-chain atom. The ridges caused by the side chains separated by four residues are shown as lines. (c) The same as (b), but here the ridges are caused by side chains separated by three residues.

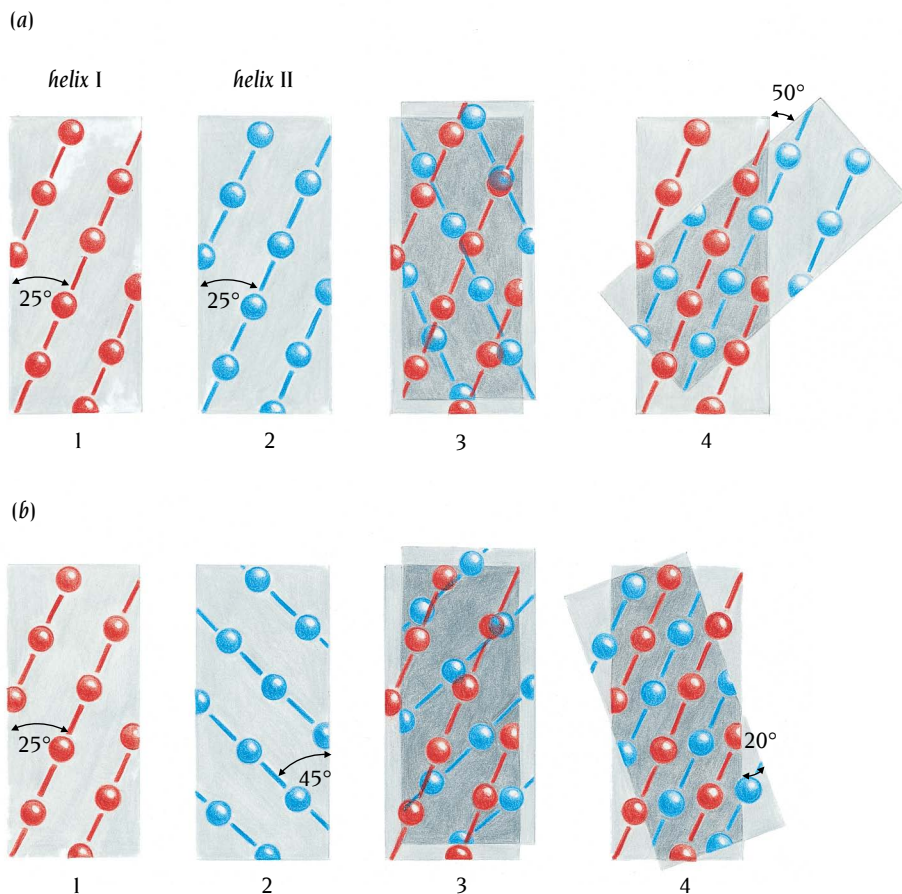
grooves form an angle of about  $25^\circ$  to the helical axis. In order to pack the two helices shown in Figure 3.12a (red and blue) against each other, one of these (the blue in Figure 3.12a) must be turned around  $180^\circ$  out of the plane of the paper and placed on top of the other (red). In the interface between the two  $\alpha$  helices the directions of the ridges and grooves are then on opposite sides of the vertical axis, as illustrated in Figure 3.12a. The  $\alpha$  helices must thus be inclined by an angle of about  $50^\circ$  ( $25^\circ + 25^\circ$ ) in order for the ridges of one helix to fit into the grooves of the other and vice versa. This is the type of packing of several of the helix-helix interactions in the globin fold, and in many other helical structures.

In the second frequently occurring packing mode the ridges formed by amino acids three residues apart fit into the grooves of amino acids four residues apart and vice versa. The direction of the first type of ridge forms an angle of about  $45^\circ$  to the helical axis, whereas the other type makes an angle of about  $25^\circ$  to the axis in the opposite direction (Figure 3.12b). In the interface, however, after one helix has been rotated  $180^\circ$ , these directions are on the same side of the helical axis. Thus an inclination of about  $20^\circ$  ( $45^\circ - 25^\circ$ ) between the two  $\alpha$  helices will fit these ridges and grooves into each other. Some four-helix-bundle structures (see Figure 3.8b) have this mode of packing.

These two rules for fitting ridges into grooves are quite general: they apply to most packing interactions between  $\alpha$  helices, and they explain the geometrical arrangements of adjacent  $\alpha$  helices observed in many protein structures.

### *The globin fold has been preserved during evolution*

The three-dimensional structures of globin domains from many diverse sources, including mammals, insects, and plant root nodules, have been determined independently of each other. All these domains have amino acid



**Figure 3.12** By fitting the ridges of side chains from one helix into the grooves between side chains of the other helix and vice versa,  $\alpha$  helices pack against each other. (a) Two  $\alpha$  helices, I and II, with ridges from side chains separated by four residues marked in red and blue, respectively. Panels 1 and 2 are the same view of the two  $\alpha$  helices. In panel 3 the blue  $\alpha$  helix is turned over through  $180^\circ$  in order to form an interface with the red  $\alpha$  helix. In panel 4 the orientation of the helices has been rotated  $50^\circ$  in order to pack the ridges of one  $\alpha$  helix into the grooves of the other. (b) In the red  $\alpha$  helix the ridges are formed by side chains separated by four residues and in the blue  $\alpha$  helix by three residues. The  $\alpha$  helices are rotated  $20^\circ$  in order to pack ridges into grooves, in a direction opposite that in (a). (Adapted from C. Chothia et al., *Proc. Natl. Acad. Sci. USA* 74: 4130–4134, 1977.)

sequence homologies that range from 99% to 16% in pairwise comparisons, but they all share the same essential features of the globin fold. This family of structures is thus the prime example of a situation where natural selection has produced proteins whose amino acid sequences have diverged widely (although some homology is usually still recognizable) but whose three-dimensional structure has been essentially preserved.

Arthur Lesk and Cyrus Chothia at the MRC Laboratory of Molecular Biology in Cambridge, UK, compared the family of globin structures with the aim of answering two general questions: How can amino acid sequences that are very different form proteins that are very similar in their three-dimensional structure? What is the mechanism by which proteins adapt to mutations in the course of their evolution?

### *The hydrophobic interior is preserved*

To answer the first question, Lesk and Chothia examined in detail residues at structurally equivalent positions that are involved in helix-heme contacts and in packing the  $\alpha$  helices against each other. After comparing the nine globin structures then known, the 59 positions they found that fulfilled these criteria were divided into 31 positions buried in the interior of the protein and 28 in contact with the heme group. These positions are the principal determinants of both the function and the three-dimensional structure of the globin family.

One might expect these positions to exhibit a higher degree of amino acid conservation and hence sequence identity than the rest of the molecule. This is not, however, the case for distantly related molecules that have low sequence identity and derive from distantly related species. The sequence identity of these residues is no greater than in the rest of the

molecule. Since the important residues involved in packing the  $\alpha$  helices are not conserved, we used to assume that the changes which have occurred compensate each other in size. This is not the case either. The volumes occupied by the 31 buried residues vary considerably between individual members. Thus neither conserved sequence nor size-compensatory mutations in the hydrophobic core are important factors in preserving three-dimensional structure during evolution. We now know that this is also true for other proteins, such as the immunoglobulins.

Lesk and Chothia did find, however, that there is a striking preferential conservation of the hydrophobic character of the amino acids at the 59 buried positions, but that no such conservation occurs at positions exposed on the surface of the molecule. With a few exceptions on the surface, hydrophobic residues have replaced hydrophilic ones and vice versa. However, the case of sickle-cell hemoglobin, which is described below, shows that a charge balance must be preserved to avoid hydrophobic patches on the surface. In summary, the evolutionary divergence of these nine globins has been constrained primarily by an almost absolute conservation of the hydrophobicity of the residues buried in the helix-to-helix and helix-to-heme contacts.

### *Helix movements accommodate interior side-chain mutations*

Lesk and Chothia also found a simple answer to the question of how proteins adapt to changes in size of buried residues. The mode of packing the  $\alpha$  helices is the same in all the globin structures: the same types of packing of ridges into grooves occur in corresponding  $\alpha$  helices in all these structures. However, the relative positions and orientations of the  $\alpha$  helices change to accommodate changes in the volume of side chains involved in the packing.

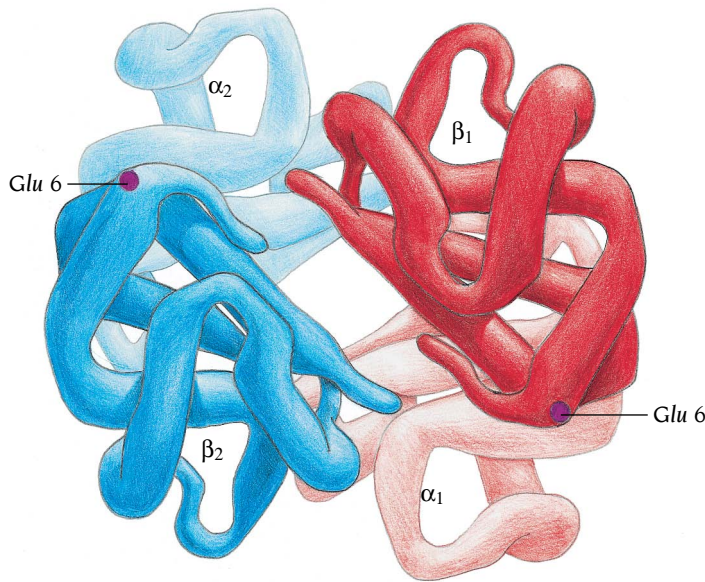
The proteins thus adapt to mutations of buried residues by changing their overall structure, which in the globins involves movements of entire  $\alpha$  helices relative to each other. The structure of loop regions changes so that the movement of one  $\alpha$  helix is not transmitted to the rest of the structure. Only movements that preserve the geometry of the heme pocket are accepted. Mutations that cause such structural shifts are tolerated because many different combinations of side chains can produce well-packed helix-helix interfaces of similar but not identical geometry and because the shifts are coupled so that the geometry of the active site is retained.

### *Sickle-cell hemoglobin confers resistance to malaria*

Sickle-cell anemia is the classic example of an inherited disease that is caused by a change in a protein's amino acid sequence. Linus Pauling proposed in 1949 that it was caused by a defect in the hemoglobin molecule; he thus coined the term **molecular disease**. Seven years later Vernon Ingram showed that the disease was caused by a single mutation, a change in residue 6 of the  $\beta$  chain of hemoglobin from Glu to Val.

Hemoglobin is a tetramer built up of two copies each of two different polypeptide chains,  $\alpha$ - and  $\beta$ -globin chains in normal adults. Each of the four chains has the globin fold with a heme pocket. Residue 6 in the  $\beta$  chain is on the surface of  $\alpha$  helix A, and it is also on the surface of the tetrameric molecule (Figure 3.13).

The hemoglobin concentration in red blood cells, erythrocytes, is extremely high, 340 mg/ml. This is almost as high as in the crystalline state: the hemoglobin molecules, which are spheroids of dimension  $50 \times 55 \times 65$  Å, are on average only 10 Å apart in the cells. It is thus surprising that they can nevertheless rotate and flow past one another. The mutation in sickle-cell hemoglobin converts a charged residue to a hydrophobic residue, and as a result, it produces a hydrophobic patch on the surface. This patch happens to fit and bind a hydrophobic pocket in the deoxygenated form of another

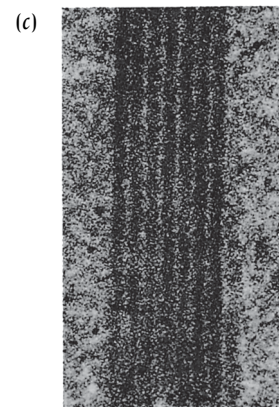
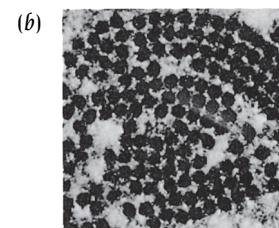
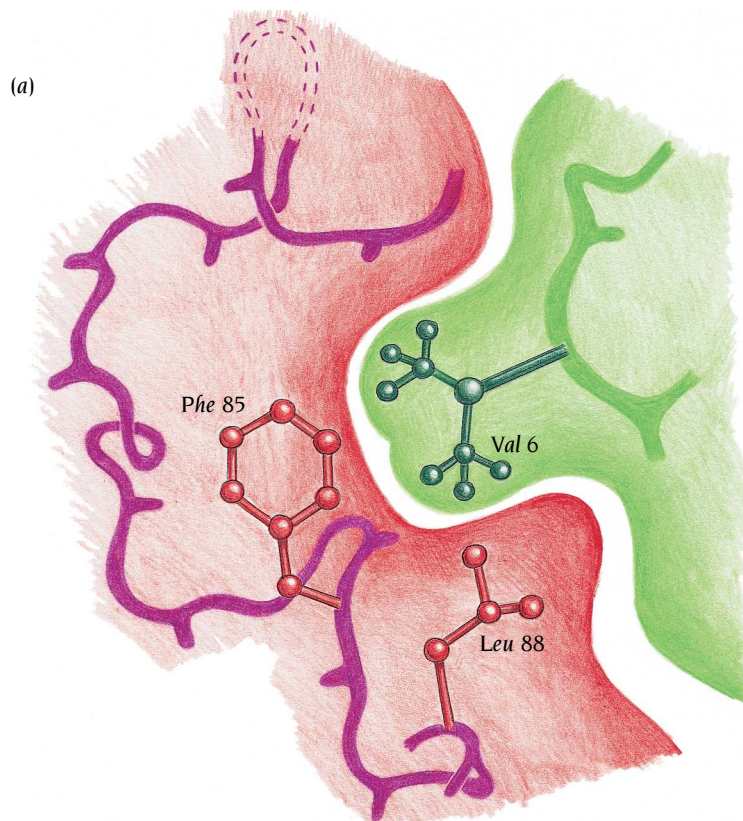


**Figure 3.13** The hemoglobin molecule is built up of four polypeptide chains: two  $\alpha$  chains and two  $\beta$  chains. Compare this with Figure 1.1 and note that for purposes of clarity parts of the  $\alpha$  chains are not shown here. Each chain has a three-dimensional structure similar to that of myoglobin: the globin fold. In sickle-cell hemoglobin Glu 6 in the  $\beta$  chain is mutated to Val, thereby creating a hydrophobic patch on the surface of the molecule. The structure of hemoglobin was determined in 1968 to 2.8 Å resolution in the laboratory of Max Perutz at the MRC Laboratory of Molecular Biology, Cambridge, UK.

hemoglobin molecule (Figure 3.14a). In the oxygenated form of hemoglobin the shape of this pocket is slightly different, and thus no interaction occurs between hemoglobin molecules in the lungs. However, when hemoglobin in the blood capillaries delivers its oxygen, these hydrophobic interactions can occur and the highly concentrated hemoglobin in the cells polymerizes into fibers (Figure 3.14b and c). These fibers stiffen the erythrocytes and deform them into a sickle shape: hence the name sickle-cell anemia. In heterozygotes, where only one  $\beta$ -globin allele is mutated, this occurs only to a minor extent, whereas it is lethal for homozygotes, all of whose hemoglobin molecules carry the mutation.

We thus have here a case where a mutation on the surface of the globin fold, replacing a hydrophilic residue with a hydrophobic one, changes important properties of the molecule and produces a lethal disease. Why has the

**Figure 3.14** Sickle-cell hemoglobin molecules polymerize due to the hydrophobic patch introduced by the mutation Glu 6 to Val in the  $\beta$  chain. The diagram (a) illustrates how this hydrophobic patch (green) interacts with a hydrophobic pocket (red) in a second hemoglobin molecule, whose hydrophobic patch interacts with the pocket in a third molecule, and so on. Electron micrographs of sickle-cell hemoglobin fibers are shown in cross-section in (b) and along the fibers in (c). [(b) and (c) from J.T. Finch et al., *Proc. Natl. Acad. Sci. USA* 70: 718–722, 1973.]



mutation survived during evolution? It turns out that the disease gives increased resistance to malaria which has had a high survival value for heterozygotes, especially in Africa. In evolutionary terms, the death of homozygotes has been an acceptable price to pay for increased survival of heterozygotes in a malarial environment.

## Conclusion

Coiled-coil  $\alpha$ -helical structures are found both in fibrous proteins and as parts of smaller domains in many globular proteins. Alpha- ( $\alpha$ -) domain structures consist of a bundle of  $\alpha$  helices that are packed together to form a hydrophobic core. A common motif is the four-helix bundle structure, where four helices are pairwise arranged in either a parallel or an antiparallel fashion and packed against each other. The most intensively studied  $\alpha$  structure is the globin fold, which has been found in a large group of related proteins, including myoglobin and hemoglobin. This structure comprises eight  $\alpha$  helices that wrap around the core in different directions and form a pocket where the heme group is bound.

Rules have been derived that explain the different geometrical arrangements of  $\alpha$  helices observed in  $\alpha$ -domain structures. The helix packing in coiled-coil structures is determined by fitting the knobs of side chains in the first helix into holes between side chains in the second helix. For other  $\alpha$ -helical structures the helix packing is determined by fitting ridges of side chains along one  $\alpha$  helix into grooves between side chains of another helix.

The globin fold has been used to study evolutionary constraints for maintaining structure and function. Evolutionary divergence is primarily constrained by conservation of the hydrophobicity of buried residues. In contrast, neither conserved sequence nor size-compensatory mutations in the hydrophobic core are important. Proteins adapt to mutations in buried residues by small changes of overall structure that in the globins involve movements of entire helices relative to each other.

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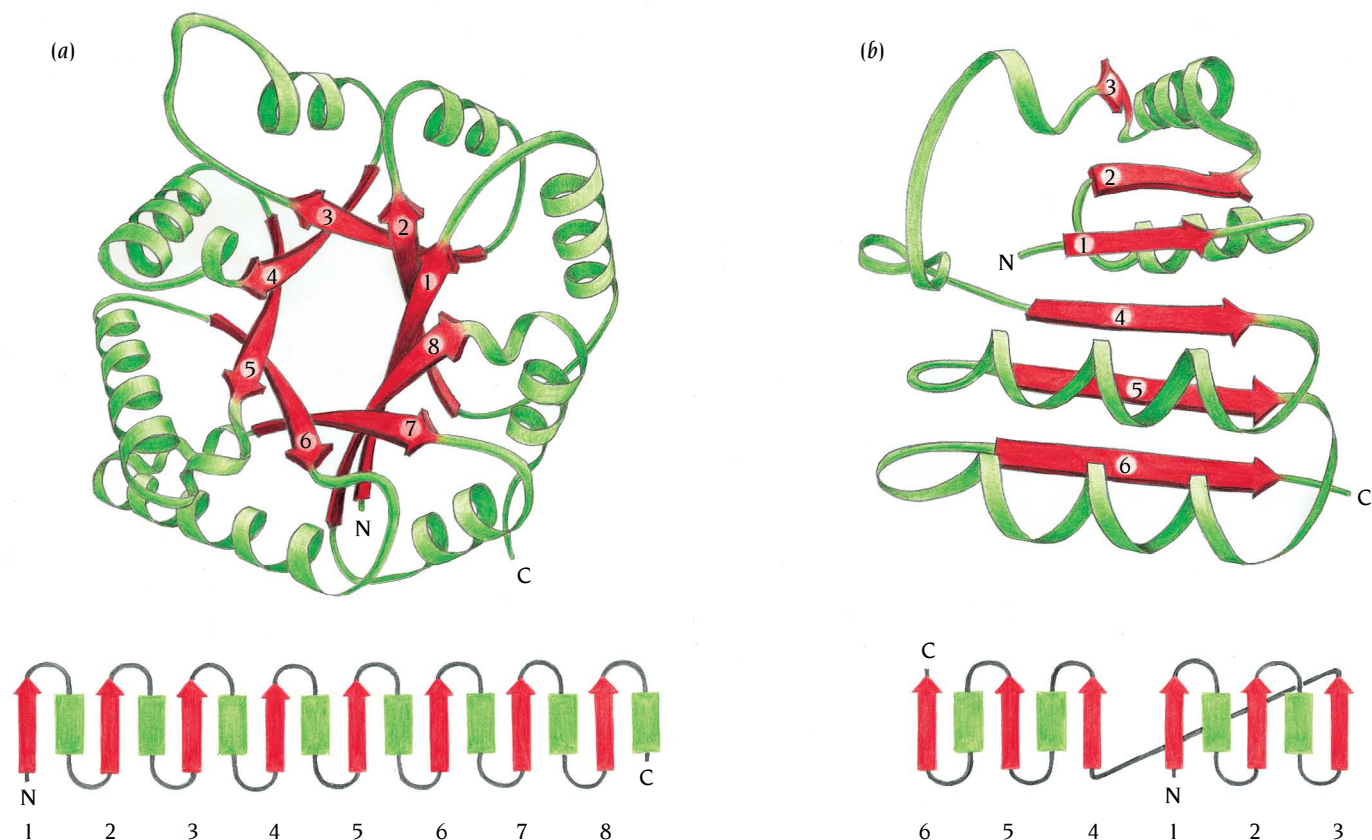
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The most frequent of the domain structures are the alpha/beta ( $\alpha/\beta$ ) domains, which consist of a central parallel or mixed  $\beta$  sheet surrounded by  $\alpha$  helices. All the glycolytic enzymes are  $\alpha/\beta$  structures as are many other enzymes as well as proteins that bind and transport metabolites. In  $\alpha/\beta$  domains, binding crevices are formed by loop regions. These regions do not contribute to the structural stability of the fold but participate in binding and catalytic action.

## *Parallel $\beta$ strands are arranged in barrels or sheets*

There are three main classes of  $\alpha/\beta$  proteins. In the first class there is a core of twisted parallel  $\beta$  strands arranged close together, like the staves of a barrel. The  $\alpha$  helices that connect the parallel  $\beta$  strands are on the outside of this barrel (Figure 4.1a). This domain structure is often called the **TIM barrel** from the structure of the enzyme triosephosphate isomerase, where it was first observed. The second class contains an open twisted  $\beta$  sheet surrounded by  $\alpha$  helices on both sides. A typical example is shown in Figure 4.1b, a nucleotide-binding domain sometimes called the **Rossmann fold** after Michael Rossmann, Purdue University, who first discovered this fold in the enzyme lactate dehydrogenase in 1970. The third class is formed by amino acid sequences that contain repetitive regions of a specific pattern of leucine residues, so-called **leucine-rich motifs**, which form  $\alpha$  helices and  $\beta$  strands. The  $\beta$  strands form a curved parallel  $\beta$  sheet with all the  $\alpha$  helices on the outside. The structure of one member of this class, a ribonuclease inhibitor (illustrated in Figure 4.11), is shaped like a horseshoe, and consequently this class is called the **horseshoe fold**.

Barrels, open sheets, and horseshoe structures are all built up from  $\beta$ - $\alpha$ - $\beta$  motifs. To illustrate how they differ, let us consider two  $\beta$ - $\alpha$ - $\beta$  motifs:  $\beta_1$ - $\alpha_{1,2}$ - $\beta_2$  and  $\beta_3$ - $\alpha_{3,4}$ - $\beta_4$  linked together by helix  $\alpha_{2,3}$ . There are two fundamentally different ways these two motifs can be connected into a  $\beta$  sheet of four parallel strands, as shown in Figure 4.2. Strand  $\beta_3$  can be aligned adjacent either to strand  $\beta_2$ , giving the strand order 1 2 3 4, or to strand  $\beta_1$ , giving the strand order 4 3 1 2. In the first case the two  $\beta$ - $\alpha$ - $\beta$  motifs are joined with the same orientation. Since the  $\beta$ - $\alpha$ - $\beta$  unit is almost always a right-handed structure, all three  $\alpha$  helices (one from each motif and the joining helix) are on the same side, above the  $\beta$  sheet (Figure 4.2a). In barrel and horseshoe structures the  $\beta$ - $\alpha$ - $\beta$  motifs are linked in this way and consist of consecutive  $\beta$ - $\alpha$ - $\beta$  units, all in the same orientation.



**Figure 4.1** Alpha/beta domains are found in many proteins. They occur in different classes, two of which are shown here: (a) a closed barrel exemplified by schematic and topological diagrams of the enzyme triosephosphate isomerase and (b) an open twisted sheet with helices on both sides, as in the coenzyme-binding domain of some dehydrogenases. Both classes are built up from  $\beta$ - $\alpha$ - $\beta$  motifs that are linked such that the  $\beta$  strands are parallel. Rectangles represent  $\alpha$  helices, and arrows represent  $\beta$  strands in the topological diagrams. [(a) Adapted from J. Richardson. (b) Adapted from B. Furugren.]

In the second case we must turn the second motif around in order to align  $\beta$  strands 1 and 3. As a result of the right-handed structure of the  $\beta$ - $\alpha$ - $\beta$  motif, its  $\alpha$  helix is on the other side of the  $\beta$  sheet (Figure 4.2b). In open twisted  $\beta$ -sheet structures there are always one or more such alignments and therefore there are  $\alpha$  helices on both sides of the  $\beta$  sheet. These geometric rules apply because virtually all  $\beta$ - $\alpha$ - $\beta$  motifs are right-handed. As pointed out in Chapter 2, this is an empirical rule that almost always applies, although no convincing explanation has been found.

### Alpha/beta barrels occur in many different enzymes

In  $\alpha/\beta$  structures where the strand order is 1 2 3 4, all connections are on the same side of the  $\beta$  sheet. An open twisted  $\beta$  sheet of this sort with four or more parallel  $\beta$  strands would leave one side of the parallel  $\beta$  sheet exposed to the solvent and the other side shielded by the  $\alpha$  helices. Such a domain structure is rarely observed, except in the horseshoe structure or as part of more complex structures where loop regions, extra  $\alpha$  helices, or additional  $\beta$  sheets cover the exposed side of the  $\beta$  sheet. Instead, a closed barrel of twisted  $\beta$  strands is formed with all the connecting  $\alpha$  helices on the outside of the barrel, as shown in Figure 4.1a. However, more than four  $\beta$  strands are needed to provide enough staves to form a closed barrel, and almost all the closed  $\alpha/\beta$  barrels observed to date have eight parallel  $\beta$  strands. These are arranged such that  $\beta$  strand 8 is adjacent and hydrogen-bonded to  $\beta$  strand 1. In a few cases the barrels do not have eight parallel  $\beta$  strands; there are also barrels that contain ten parallel  $\beta$  strands and some that contain eight parallel and two antiparallel  $\beta$  strands. In almost all cases the cross-connections between the parallel  $\beta$  strands are  $\alpha$  helices; in addition, there is usually an  $\alpha$  helix after the last  $\beta$  strand.

The eight-stranded  $\alpha/\beta$ -barrel structure is one of the largest and most regular of all domain structures. A minimum of about 200 residues are required to form this structure. It has been found in many different proteins, most of which are enzymes, with completely different amino acid sequences and



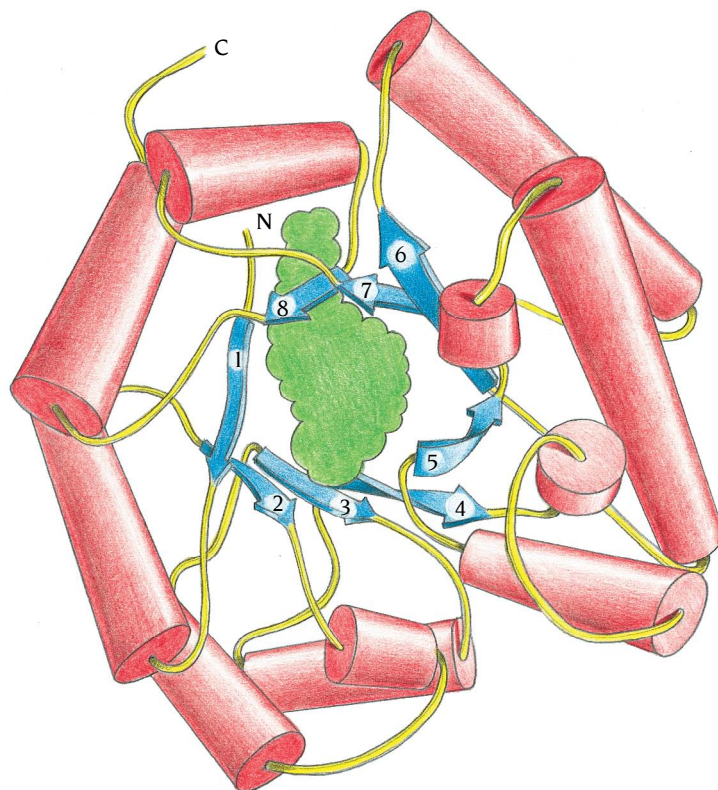
**Table 4.1** The amino acid residues of the eight parallel  $\beta$  strands in the barrel structure of the enzyme triosephosphate isomerase from chicken muscle

| Strand no. | Residue no. | Positions |     |     |     |     |
|------------|-------------|-----------|-----|-----|-----|-----|
|            |             | 1         | 2   | 3   | 4   | 5   |
| 1          | 6–10        | Phe       | Val | Gly | Gly | Asn |
| 2          | 37–41       | Glu       | Val | Val | Cys | Gly |
| 3          | 59–63       | Gly       | Val | Ala | Ala | Gln |
| 4          | 89–93       | Trp       | Val | Ile | Leu | Gly |
| 5          | 121–125     | Gly       | Val | Ile | Ala | Cys |
| 6          | 158–162     | Lys       | Val | Val | Leu | Ala |
| 7          | 204–208     | Arg       | Ile | Ile | Tyr | Gly |
| 8          | 227–231     | Gly       | Phe | Leu | Val | Gly |

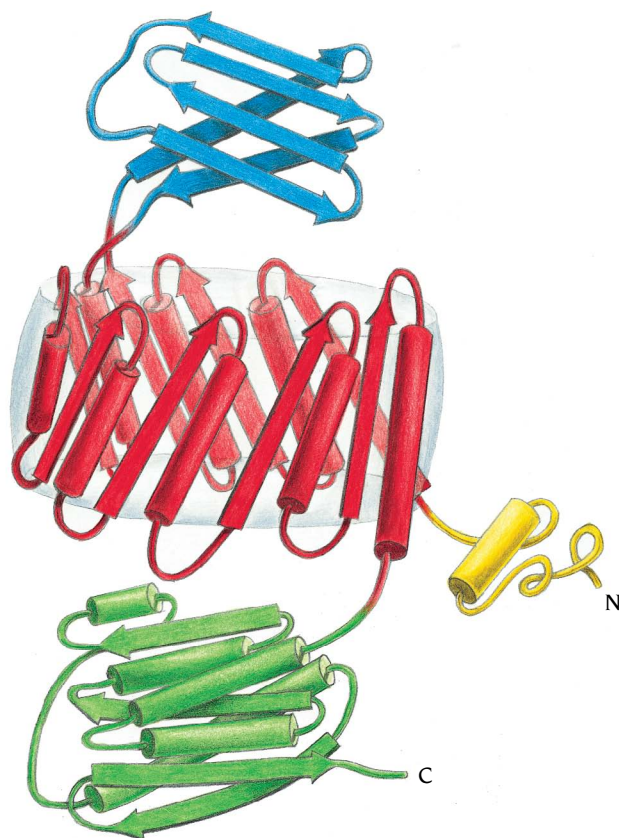
The sequences are aligned so that residues in positions 1, 3, and 5 point into the barrel and residues in positions 2 and 4 point toward the  $\alpha$  helices on the outside and are involved in the hydrophobic interactions between the  $\beta$  strands and the  $\alpha$  helices.

Bulky hydrophobic residues from positions 1, 3, and 5 of the  $\beta$  strands fill the interior of the barrel and form a tightly packed hydrophobic core (see Figure 4.3). Note from Table 4.1 that some of these residues are Lys, Arg, or Gln, which have a polar end group (see Panel 1.1, pp. 6–7) terminating a chain of hydrophobic  $-\text{CH}_2$  groups. These chains are in the hydrophobic interior and traverse part of the barrel; their polar end groups are on the top or bottom surface of the barrel and are in contact with the aqueous environment. By this arrangement even amino acids that are classified as polar can participate in the formation of hydrophobic cores of compact globular domains through the hydrophobic parts of their side chains.

There is one exception to the rule that requires bulky hydrophobic residues to fill the interior of eight-stranded  $\alpha/\beta$  barrels in order to form a tightly packed hydrophobic core. The coenzyme  $\text{B}_{12}$ -dependent enzyme methylmalonyl-coenzyme A mutase, the x-ray structure of which was determined by Phil Evans and colleagues at the MRC Laboratory of Molecular



**Figure 4.4** Schematic diagram of the structure of the  $\alpha/\beta$ -barrel domain of the enzyme methylmalonyl-coenzyme A mutase. Alpha helices are red, and  $\beta$  strands are blue. The inside of the barrel is lined by small hydrophilic side chains (serine and threonine) from the  $\beta$  strands, which creates a hole in the middle where one of the substrate molecules, coenzyme A (green), binds along the axis of the barrel from one end to the other. (Adapted from a computer-generated diagram provided by P. Evans.)



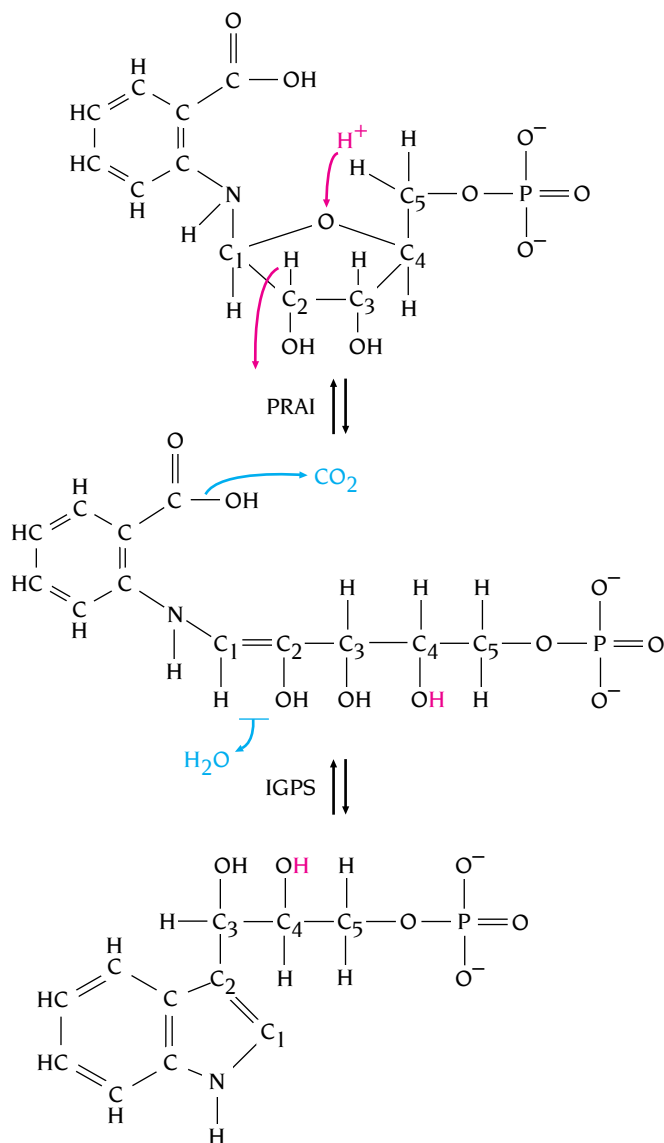
**Figure 4.5** The polypeptide chain of the enzyme pyruvate kinase folds into several domains, one of which is an  $\alpha/\beta$  barrel (red). One of the loop regions in this barrel domain is extended and comprises about 100 amino acid residues that fold into a separate domain (blue) built up from antiparallel  $\beta$  strands. The C-terminal region of about 140 residues forms a third domain (green), which is an open twisted  $\alpha/\beta$  structure.

Biology, in Cambridge, UK, has a hole in the middle of its  $\alpha/\beta$ -barrel domain (Figure 4.4). The serine and threonine side chains that project from the  $\beta$  strands into the interior of the barrel are small and polar, and therefore do not fill up the space available inside the barrel. The resulting tunnel through the barrel provides an ideal environment for the catalytic reaction and is sufficiently large for the substrate molecule, methylmalonyl-coenzyme A, to bind. Many enzyme-catalyzed reactions, including the reaction catalyzed by this mutase, require that the reactive part of the substrate molecule be shielded from solvent during the catalytic reaction. When the substrate is bound in the tunnel inside the barrel, it is shielded from the outside world. In  $\alpha/\beta$  barrels with a hydrophobic core, the substrate binds at the surface of the barrel and conformational changes of loop regions shield the substrate from the solvent.

### *Pyruvate kinase contains several domains, one of which is an $\alpha/\beta$ barrel*

All known eight-stranded  $\alpha/\beta$ -barrel domains have enzymatic functions that include isomerization of small sugar molecules, oxidation by flavin co-enzymes, phosphate transfer, and degradation of sugar polymers. In some of these enzymes the barrel domain comprises the whole subunit of the protein; in others the polypeptide chain is longer and forms several additional domains. An enzymatic function in these multidomain subunits, however, is always associated with the barrel domain.

For example, each subunit of the dimeric glycolytic enzyme triosephosphate isomerase (see Figure 4.1a) consists of one such barrel domain. The polypeptide chain has 248 residues in which the first  $\beta$  strand of the barrel starts at residue 6 and the last  $\alpha$  helix of the barrel ends at residue 246. In contrast, the subunit of the glycolytic enzyme pyruvate kinase (Figure 4.5), which was solved at 2.6 Å resolution in the laboratory of Hilary Muirhead, Bristol University, UK, is folded into four different domains. The polypeptide chain of this cat muscle enzyme has 530 residues. In Figure 4.5, residues 1–42



**Figure 4.6** The bifunctional enzyme PRA-isomerase (PRAI):IGP-synthase (IGPS) catalyzes two sequential reactions in the biosynthesis of tryptophan. In the first reaction (top half), which is catalyzed by the C-terminal PRAI domain of the enzyme, the substrate N-(5'-phosphoribosyl) anthranilate (PRA) is converted to 1-(o-carboxyphenylamino)-1-deoxyribulose 5-phosphate (CdRP) by a rearrangement reaction. The succeeding step (bottom half), a ring closure reaction from CdRP to indole-3-glycerol phosphate (IGP), is catalyzed by the N-terminal IGPS domain.

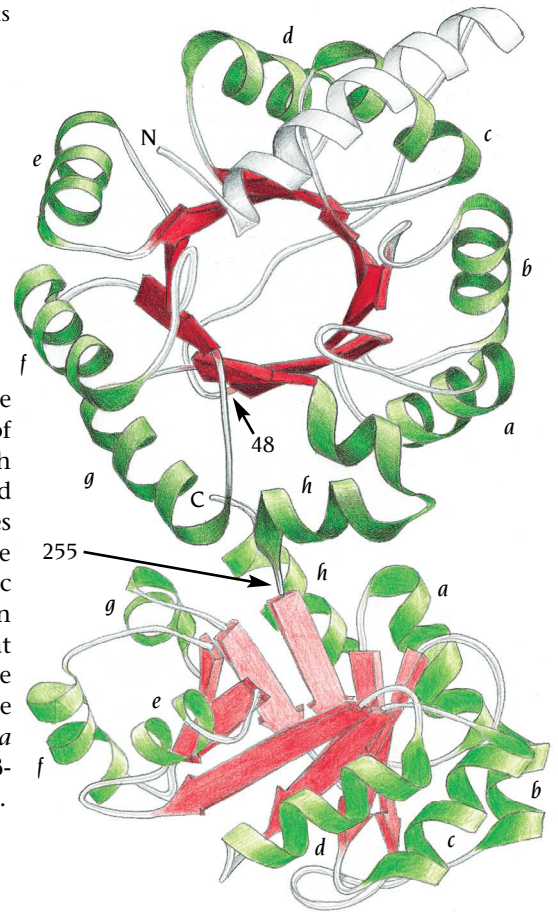
form a small domain (yellow) involved in subunit contacts in the tetrameric molecule; residues 43–115 and 224–387 form an  $\alpha/\beta$ -barrel domain (red) that binds substrate and provides the catalytic groups; residues 116–223 loop out from the end of  $\beta$ -strand number 3 in the barrel domain and are folded into a separate domain consisting of an antiparallel  $\beta$  sheet (blue); and finally, residues 388–530 form an open twisted  $\alpha/\beta$  domain (green). This structure illustrates perfectly how a long polypeptide chain can be arranged in domains of different structural types.

### Double barrels have occurred by gene fusion

PRA-isomerase:IGP-synthase, a bifunctional enzyme from *E. coli* that catalyzes two reactions in the synthesis of tryptophan (Figure 4.6), has a polypeptide chain that forms two  $\alpha/\beta$  barrels. The structure of this enzyme, solved at 2.8 Å in the laboratory of Hans Jansonius in Basel, Switzerland, showed that residues 48–254 form one barrel with IGP-synthase activity, while residues 255–450 form the second barrel with PRA-isomerase activity (Figure 4.7).

In *Bacillus subtilis* these two reactions are catalyzed by two separate enzymes that have amino acid sequences homologous to the corresponding regions of the bifunctional enzyme from *E. coli*, and thus each forms a barrel

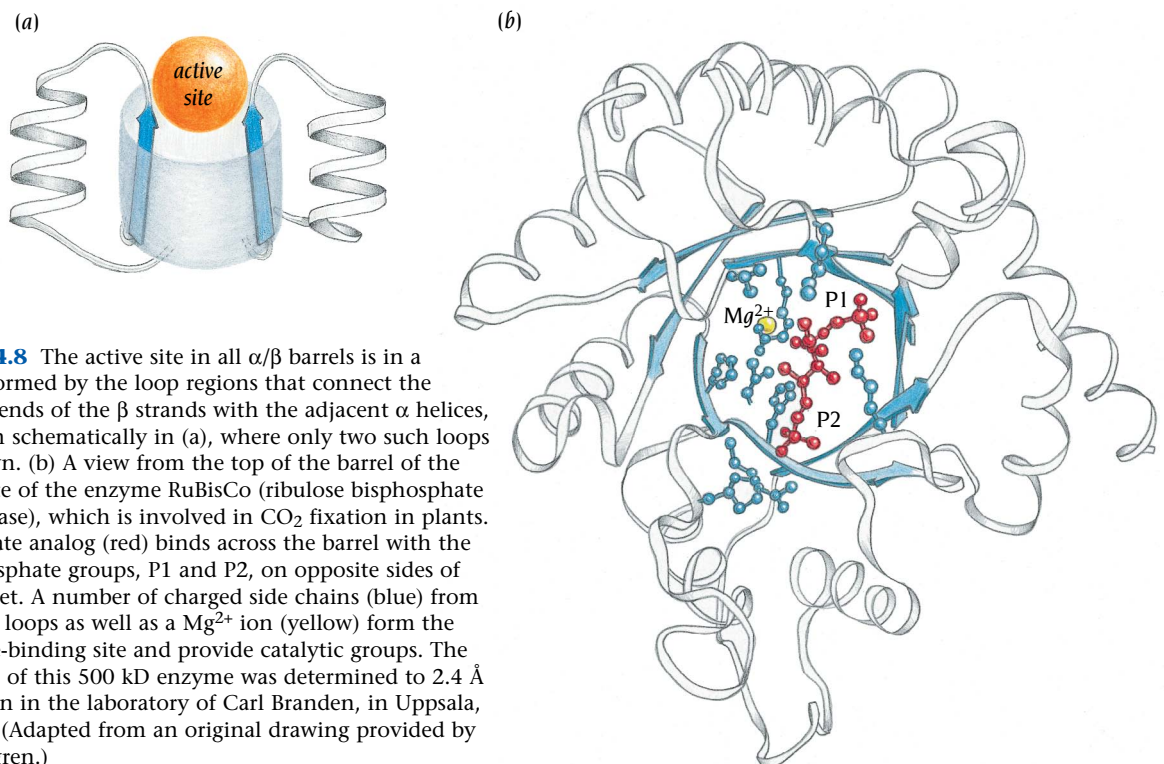
**Figure 4.7** Two of the enzymatic activities involved in the biosynthesis of tryptophan in *E. coli*, phosphoribosyl anthranilate (PRA) isomerase and indoleglycerol phosphate (IGP) synthase, are performed by two separate domains in the polypeptide chain of a bifunctional enzyme. Both these domains are  $\alpha/\beta$ -barrel structures, oriented such that their active sites are on opposite sides of the molecule. The two catalytic reactions are therefore independent of each other. The diagram shows the IGP-synthase domain (residues 48–254) with dark colors and the PRA-isomerase domain with light colors. The  $\alpha$  helices are sequentially labeled a–h in both barrel domains. Residue 255 (arrow) is the first residue of the second domain. (Adapted from J.P. Priestle et al., *Proc. Natl. Acad. Sci. USA* 84: 5690–5694, 1987.)



structure. There is no obvious functional advantage to *E. coli* in having these two enzymatic activities in one polypeptide chain, since the active sites of the two barrels are on opposite sides of the molecule facing away from each other and the two reactions are thus independent of each other. A third organism, *Neurospora crassa*, has an enzyme with three catalytic activities within the same polypeptide chain; here two domains similar to those of the *E. coli* enzyme are linked to a third domain that has yet another enzymatic function in the same biosynthetic pathway. These differences between species reflect different ways to organize the genome. DNA sequences that code for protein domains with different functions are organized into separate genes in one organism and fused into a single gene in another. Although the three-dimensional structures of these enzymes in *B. subtilis* and *N. crassa* have not been solved by crystallography, we can be certain that they are  $\alpha/\beta$ -barrel domains because of their sequence homologies to the *E. coli* proteins.

### The active site is formed by loops at one end of the $\alpha/\beta$ barrel

In all these  $\alpha/\beta$ -barrel domains the active site is in a very similar position. It is situated in the bottom of a funnel-shaped pocket created by the eight loops that connect the carboxy end of the  $\beta$  strands with the amino end of the  $\alpha$  helices (Figure 4.8). Residues that participate in binding and catalysis are in



**Figure 4.8** The active site in all  $\alpha/\beta$  barrels is in a pocket formed by the loop regions that connect the carboxy ends of the  $\beta$  strands with the adjacent  $\alpha$  helices, as shown schematically in (a), where only two such loops are shown. (b) A view from the top of the barrel of the active site of the enzyme RuBisCo (ribulose biphosphate carboxylase), which is involved in  $\text{CO}_2$  fixation in plants. A substrate analog (red) binds across the barrel with the two phosphate groups, P1 and P2, on opposite sides of the pocket. A number of charged side chains (blue) from different loops as well as a  $\text{Mg}^{2+}$  ion (yellow) form the substrate-binding site and provide catalytic groups. The structure of this 500 kD enzyme was determined to 2.4 Å resolution in the laboratory of Carl Branden, in Uppsala, Sweden. (Adapted from an original drawing provided by Bo Furugren.)

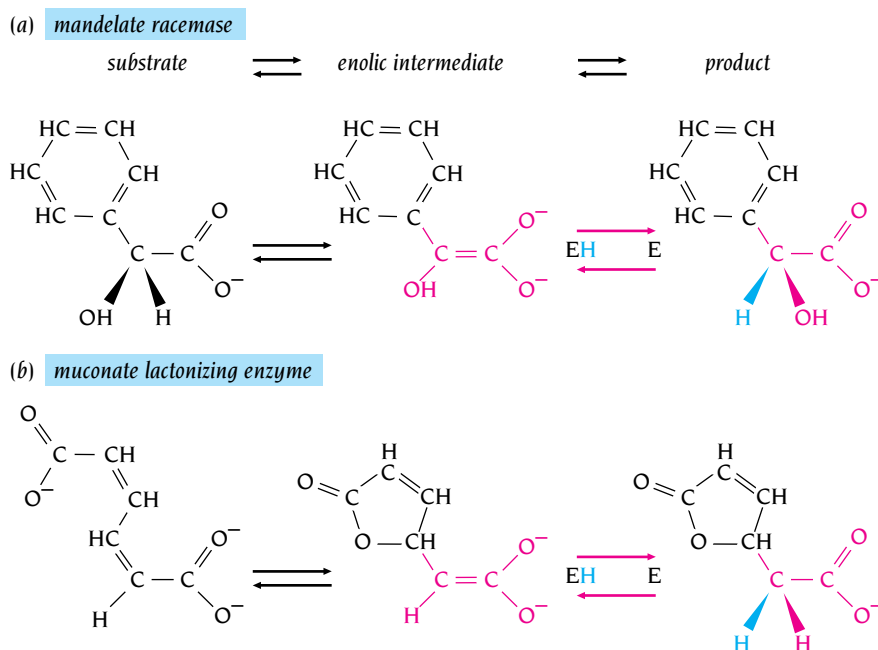
these loop regions. In other words, these enzymes are modeled on a common stable scaffold of eight parallel  $\beta$  strands surrounded by eight  $\alpha$  helices. In each case the specific enzymatic activity is determined by the eight loop regions at the carboxy end of the  $\beta$  strands, which do not contribute to the structural stability of the scaffold. In some cases an additional loop region from a second domain or a different subunit comes close to this active site and also participates in binding and catalysis.

### *Alpha/beta barrels provide examples of evolution of new enzyme activities*

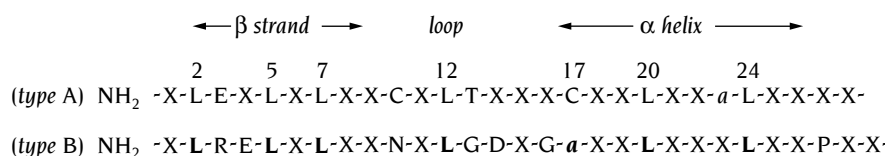
How do new enzyme activities evolve? Are new enzymes formed from random sequences generated by recombination and other genetic rearrangements or do they arise by divergent evolution from a preexisting set of enzymes. Greg Petsko at Brandeis University has provided strong evidence for the latter case from studies of  $\alpha/\beta$ -barrel enzymes in a rare metabolic pathway, conversion of mandelate to benzoate. This rare metabolic pathway is thought to be of recent evolutionary origin, since it is present in only a few pseudomonad species.

The first enzyme in this pathway, mandelate racemase, catalyzes the interconversion of the two optical isomers of mandelate (Figure 4.9a). The key step in this reaction is proton abstraction from a carbon atom, producing an enolic intermediate. Petsko found that the three-dimensional structure of this enzyme, including its  $\alpha/\beta$  barrel, is very similar to that of a quite different enzyme, muconate lactonizing enzyme, which catalyzes a different chemical reaction (Figure 4.9b) but which also involves the formation of an intermediate by proton abstraction. The amino acid sequences of the 350 residues of these enzymes showed 26% sequence identity, which clearly demonstrates that they are evolutionarily related. By comparing these two structures in detail Petsko found significant similarities in the region of the active site that catalyzes proton abstraction and intermediate formation but substantial differences in those regions of the active site that confer substrate specificity.

These results are compatible with an evolutionary history in which the new enzyme activity of mandelate racemase has evolved from a preexisting enzyme that catalyzes the basic chemical reaction of proton abstraction and formation of an intermediate. Subsequent mutations have modified the



**Figure 4.9** Mechanisms of the reactions catalyzed by the enzymes mandelate racemase (a) and muconate lactonizing enzyme (b). The two overall reactions are quite different: a change of configuration of a carbon atom for mandelate racemase versus ring closure for the lactonizing enzyme. However, one crucial step (pink) in the two reactions is the same: addition of a proton (blue) to an intermediate of the substrate (pink) from a lysine residue of the enzyme (E) or, in the reverse direction, formation of an intermediate by proton abstraction from the carbon atom adjacent to the carboxylate group.



**Figure 4.10** Consensus amino acid sequence and secondary structure of the leucine-rich motifs of type A and type B. “X” denotes any amino acid; “a” denotes an aliphatic amino acid. Conserved residues are shown in bold in type B.

substrate specificity while preserving the ability to catalyze the basic chemical reaction. Chemistry is the important factor to preserve during evolution of new enzymes, while specificity can be modified. It would therefore seem that relatively nonspecific enzymes, which may have existed earlier in evolution or which may arise occasionally through random genetic rearrangements, are the clay from which nature sculpts new enzymes. To preserve the original enzyme activity and at the same time allow divergence, the precursor gene for the enzyme must first be duplicated at some point.

Further strong evidence that proteins with new functions evolve by the process of gene duplication and subsequent modification by mutation has recently been found by examination of the genome sequence of the bacterium *Haemophilus influenzae*, the first free-living organism to be completely sequenced. Sequence comparisons showed that, of the 1680 identified proteins coded by this genome, at least one third are related to one or more other proteins within the genome and therefore have arisen from processes that involve gene duplications.

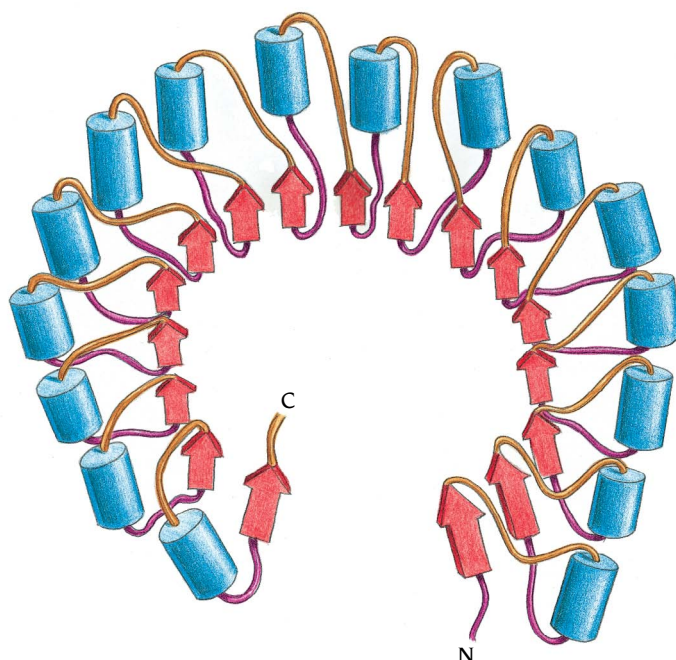
Alpha/beta barrels are particularly well suited to such an evolutionary strategy, since substrate specificity and catalytic function reside in loop regions which are separated from the residues of the  $\alpha$  helices and  $\beta$  strands that contribute to the structural stability of these domain structures. Such enzymes should also be excellent targets for genetic redesign *in vitro*. By changing the lengths and specific residues of the active-site loop regions, it might be possible to produce novel substrate specificities without affecting the stability of the structural framework and therefore the enzyme.

### Leucine-rich motifs form an $\alpha/\beta$ -horseshoe fold

Leucine-rich motifs—tandem homologous amino acid sequences of about 20–30 residues—have been identified from sequence studies in over 60 different proteins, including receptors, cell adhesion molecules, bacterial virulence factors, and molecules involved in RNA splicing and DNA repair. The x-ray structure of one member of this class of proteins, a ribonuclease inhibitor, has been determined by Johann Deisenhofer and colleagues at the University of Texas, Dallas. The 456 amino acids of the polypeptide chain are arranged in 15 tandem leucine-rich motifs of two types that alternate along the chain: type A, with 29 residues, and type B, with 28 residues. In addition there are two short regions with nonhomologous sequences at the termini of the chain. The consensus sequence of these homologous repeats (Figure 4.10) indicates that both types of repeat contain a characteristic pattern of leucine residues that play an important structural role, as we will see.

Each repeat forms a right-handed  $\beta$ -loop- $\alpha$  structure similar to those found in the two other classes of  $\alpha/\beta$  structures described earlier. Sequential  $\beta$ -loop- $\alpha$  repeats are joined together in a similar way to those in the  $\alpha/\beta$ -barrel structures. The  $\beta$  strands form a parallel  $\beta$  sheet, and all the  $\alpha$  helices are on one side of the  $\beta$  sheet. However, the  $\beta$  strands do not form a closed barrel; instead they form a curved open structure that resembles a horseshoe with  $\alpha$  helices on the outside and a  $\beta$  sheet forming the inside wall of the horseshoe (Figure 4.11). One side of the  $\beta$  sheet faces the  $\alpha$  helices and participates in a hydrophobic core between the  $\alpha$  helices and the  $\beta$  sheet; the other side of the  $\beta$  sheet is exposed to solvent, a characteristic other  $\alpha/\beta$  structures do not have.

The leucine residues in this leucine-rich motif form a hydrophobic core between the  $\beta$  sheet and the  $\alpha$  helices. Leucine residues 2, 5, and 7 (see Figure



**Figure 4.11** Schematic diagram of the structure of the ribonuclease inhibitor. The molecule, which is built up by repetitive  $\beta$ -loop- $\alpha$  motifs, resembles a horseshoe with a 17-stranded parallel  $\beta$  sheet on the inside and 16  $\alpha$  helices on the outside. The  $\beta$  sheet is light red,  $\alpha$  helices are blue, and loops that are part of the  $\beta$ -loop- $\alpha$  motifs are orange. (Adapted from B. Kobe et al., *Nature* 366: 751–756, 1993.)

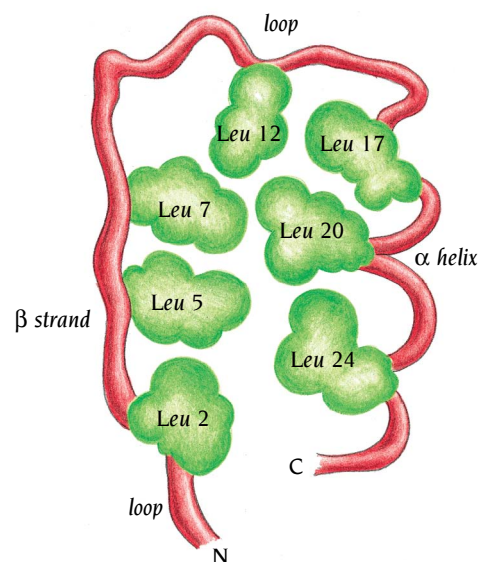
4.10) from the  $\beta$  strand of the motif pack against leucine residues 20 and 24 of the  $\alpha$  helix to form the main part of the hydrophobic region. Leucine residue 12 from the loop region is also part of this hydrophobic core, as is residue 17 from the  $\alpha$  helix, which is usually hydrophobic (Figure 4.12).

Leucine residues 2, 5, 7, 12, 20, and 24 of the motif are invariant in both type A and type B repeats of the ribonuclease inhibitor. An examination of more than 500 tandem repeats from 68 different proteins has shown that residues 20 and 24 can be other hydrophobic residues, whereas the remaining four leucine residues are present in all repeats. On the basis of the crystal structure of the ribonuclease inhibitor and the important structural role of these leucine residues, it has been possible to construct plausible structural models of several other proteins with leucine-rich motifs, such as the extracellular domains of the thyrotropin and gonadotropin receptors.

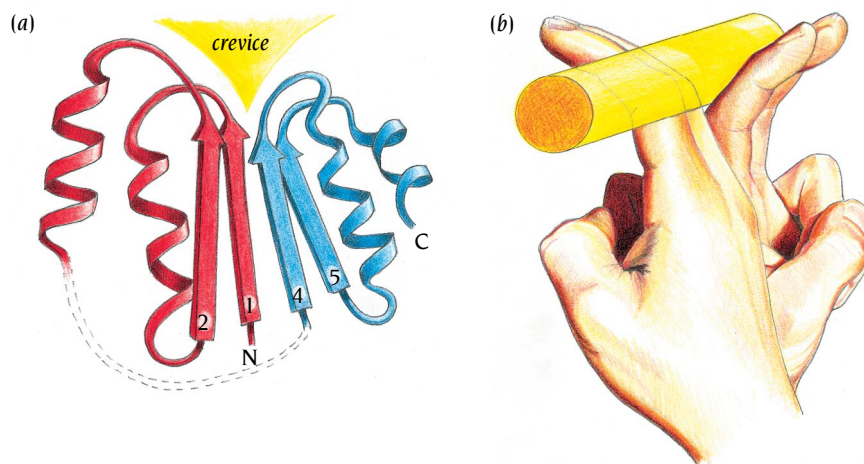
### *Alpha/beta twisted open-sheet structures contain $\alpha$ helices on both sides of the $\beta$ sheet*

In the next class of  $\alpha/\beta$  structures there are  $\alpha$  helices on both sides of the  $\beta$  sheet. This has at least three important consequences. First, a closed barrel cannot be formed unless the  $\beta$  strands completely enclose the  $\alpha$  helices on one side of the  $\beta$  sheet. Such structures have never been found and are very unlikely to occur, since a large number of  $\beta$  strands would be required to enclose even a single  $\alpha$  helix. Instead, the  $\beta$  strands are arranged into an open twisted  $\beta$  sheet such as that shown in Figure 4.1b.

Second, there are always two adjacent  $\beta$  strands ( $\beta_1$  and  $\beta_3$  in Figure 4.2b) in the interior of the  $\beta$  sheet whose connections to the flanking  $\beta$  strand are on opposite sides of the  $\beta$  sheet. One of the loops from one of these two  $\beta$  strands goes above the  $\beta$  sheet, whereas the other loop goes below. This creates a crevice outside the edge of the  $\beta$  sheet between these two loops (Figure 4.13). Almost all binding sites in this class of  $\alpha/\beta$  proteins are located in crevices of this type at the carboxy edge of the  $\beta$  sheet, as we discuss in detail



**Figure 4.12** Schematic diagram illustrating the role of the conserved leucine residues (green) in the leucine-rich motif in stabilizing the  $\beta$ -loop- $\alpha$  structural module. In the ribonuclease inhibitor, leucine residues 2, 5, and 7 from the  $\beta$  strand pack against leucine residues 17, 20, and 24 from the  $\alpha$  helix as well as leucine residue 12 from the loop to form a hydrophobic core between the  $\beta$  strand and the  $\alpha$  helix.



**Figure 4.13** (a) The active site in open twisted  $\alpha/\beta$  domains is in a crevice outside the carboxy ends of the  $\beta$  strands. This crevice is formed by two adjacent loop regions that connect the two strands with  $\alpha$  helices on opposite sides of the  $\beta$  sheet. This is illustrated by the curled fingers of two hands (b), where the top halves of the fingers represent loop regions and the bottom halves represent the  $\beta$  strands. The rod represents a bound molecule in the binding crevice.

later. (We define the carboxy edge of the sheet as the edge that is formed by the carboxy ends of the parallel  $\beta$  strands in the sheet.)

Third, in open-sheet structures the  $\alpha$  helices are packed against both sides of the  $\beta$  sheet. Each  $\beta$  strand thus contributes hydrophobic side chains to pack against  $\alpha$  helices in two similar hydrophobic core regions, one on each side of the  $\beta$  sheet.

### *Open $\beta$ -sheet structures have a variety of topologies*

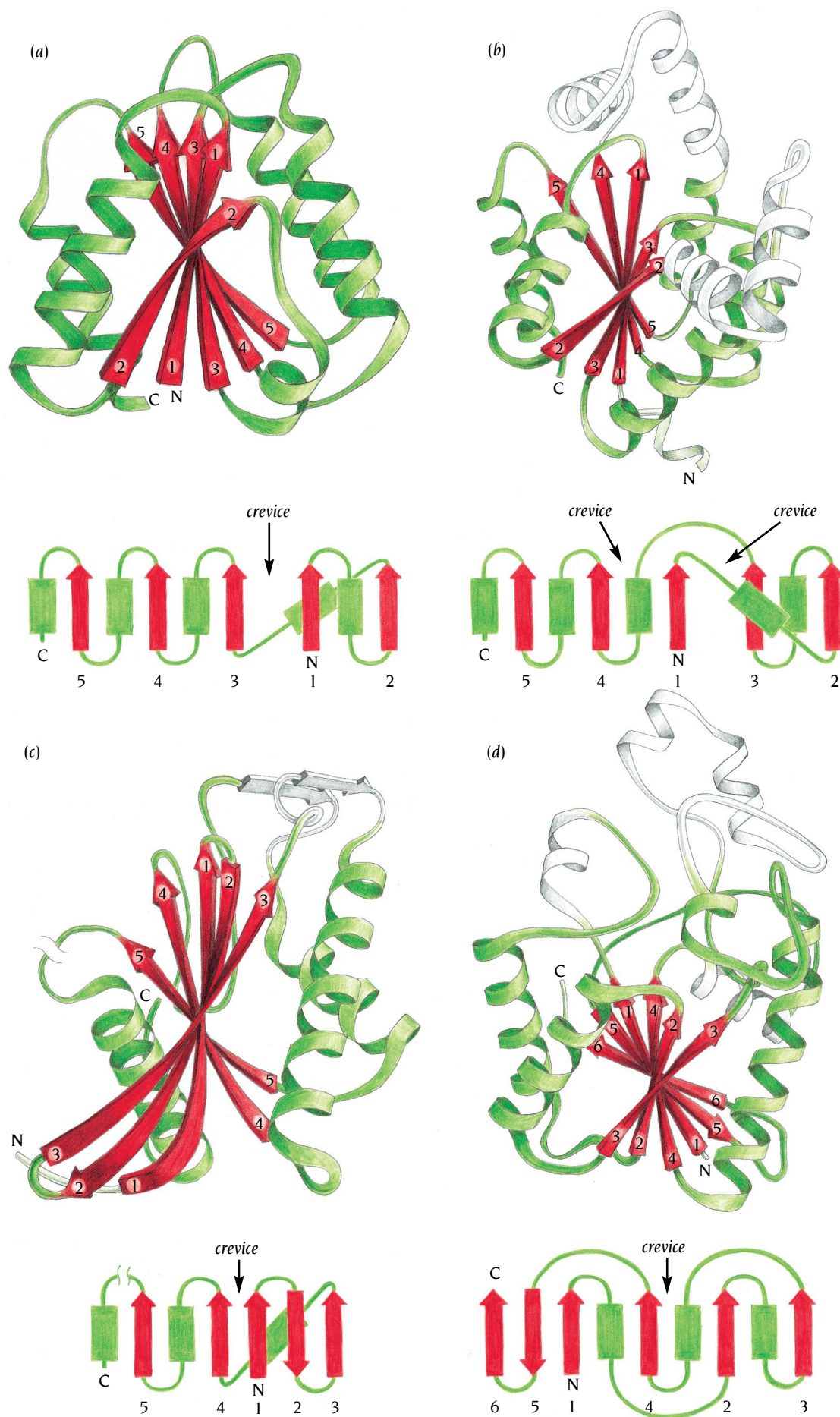
We have seen that all members of the  $\alpha/\beta$ -barrel domain structures have the same basic arrangement of eight  $\alpha$  helices and eight  $\beta$  strands. Within the open  $\alpha/\beta$  sheets, however, there is much more variation in structure, as is obvious from purely geometric considerations. Since the  $\beta$  strands form an open  $\beta$  sheet, there are no geometric restrictions on the number of strands involved. In fact, the number varies from four to ten. Furthermore, the two  $\beta$  strands joined by a crossover connection need not be adjacent in the  $\beta$  sheet, although the  $\beta$ - $\alpha$ - $\beta$  motif where the two  $\beta$  strands are adjacent is a preferred structural building block. In addition, there can be mixed  $\beta$  sheets in which hairpin connections give rise to some antiparallel  $\beta$  strands mixed with the parallel  $\beta$  strands. All these variations occur in actual structures, some of which are illustrated in Figure 4.14a–d. There are thus many variations on the regular arrangement of six parallel  $\beta$  strands (see Figure 4.1b).

### *The positions of active sites can be predicted in $\alpha/\beta$ structures*

We have described a general relationship between structure and function for the  $\alpha/\beta$ -barrel structures. They all have the active site at the same position with respect to their common structure in spite of having different functions as well as different amino acid sequences. We can now ask if similar relationships also occur for the open  $\alpha/\beta$ -sheet structures in spite of their much greater variation in structure. Can the position of the active sites be predicted from the structures of many open-sheet  $\alpha/\beta$  proteins?

In almost every one of the more than 100 different known  $\alpha/\beta$  structures of this class the active site is at the carboxy edge of the  $\beta$  sheet. Functional residues are provided by the loop regions that connect the carboxy end of the  $\beta$  strands with the amino end of the  $\alpha$  helices. In this one respect a fundamental similarity therefore exists between the  $\alpha/\beta$ -barrel structures and the open  $\alpha/\beta$ -sheet structures.

The general shapes of the active sites are quite different, however. Open  $\alpha/\beta$  structures cannot form funnel-shaped active sites like the barrel structures. Instead, they form crevices at the edge of the  $\beta$  sheet. Such crevices occur when there are two adjacent connections that are on opposite sides of the  $\beta$  sheet. One of the loop regions in these two connections goes out from



**Figure 4.14** Examples of different types of open twisted  $\alpha/\beta$  structures. Both schematic and topological diagrams are given. In the topological diagrams, arrows denote strands of  $\beta$  sheet and rectangles denote  $\alpha$  helices. (a) The FMN-binding redox protein flavodoxin. (b) The enzyme adenylate kinase, which catalyzes the reaction  $\text{AMP} + \text{ATP} \rightleftharpoons 2 \text{ADP}$ . The structure was determined to 3.0 Å resolution in the laboratory of Georg Schulz in Heidelberg, Germany. (c) The ATP-binding domain of the glycolytic enzyme hexokinase, which catalyzes the phosphorylation of glucose. The structure was determined to 2.8 Å resolution in the laboratory of Tom Steitz, Yale University. (d) The glycolytic enzyme phosphoglycerate mutase, which catalyzes transfer of a phosphoryl group from carbon 3 to carbon 2 in phosphoglycerate. The structure was determined to 2.5 Å resolution in the laboratory of Herman Watson, Bristol University, UK. (Adapted from J. Richardson.)

its  $\beta$  strand above the  $\beta$  sheet and the other below, creating a crevice between them (see Figure 4.13). The active site or part of it is usually found in such a crevice. The position of such crevices is determined by the topology of the  $\beta$  sheet and can be predicted from a topology diagram. The crevices occur when the strand order is reversed, and can be easily identified in a topology diagram as the place where connections from the carboxy ends of two adjacent  $\beta$  strands go in opposite directions, one to the left and one to the right. Let us examine the first two diagrams given in Figure 4.14.

The first structure, flavodoxin (Figure 4.14a), has one such position, between strands 1 and 3. The connection from strand 1 goes to the right and that from strand 3 to the left. In the schematic diagram in Figure 4.14a we can see that the corresponding  $\alpha$  helices are on opposite sides of the  $\beta$  sheet. The loops from these two  $\beta$  strands, 1 and 3, to their respective  $\alpha$  helices form the major part of the binding cleft for the coenzyme FMN (flavin mononucleotide).

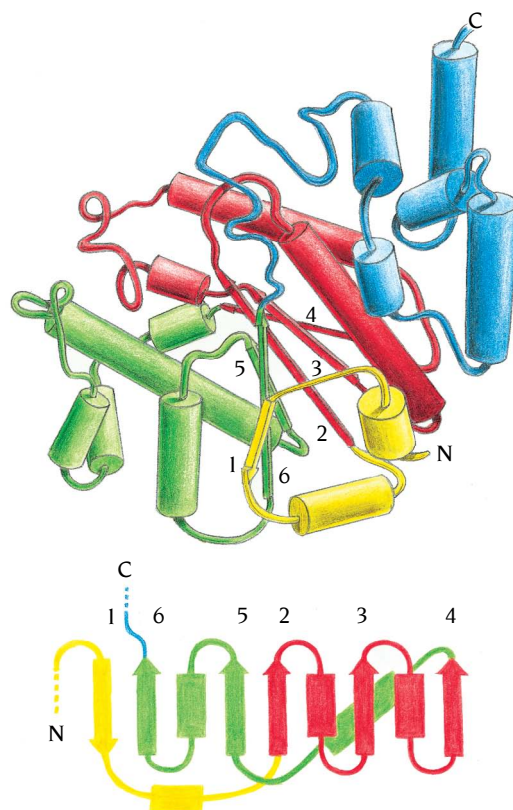
The second structure, adenylate kinase (Figure 4.14b), has two such positions, one on each side of  $\beta$  strand 1. The connection from strand 1 to strand 2 goes to the right, whereas the connection from the flanking strands 3 and 4 both go to the left. Crevices are formed between  $\beta$  strands 1 and 3 and between strands 1 and 4. One of these crevices forms part of an AMP-binding site, and the other crevice forms part of an ATP-binding site that catalyzes the formation of ADP from AMP and ATP.

Such positions in a topology diagram are called topological switch points. It was postulated in 1980 by Carl Branden, in Uppsala, Sweden, that the position of active sites could be predicted from such switch points. Since then at least one part of the active site has been found in crevices defined by such switch points in almost all new  $\alpha/\beta$  structures that have been determined. Thus we can predict the approximate position of the active site and possible loop regions that form this site in  $\alpha/\beta$  proteins. This is in contrast to proteins of the other two main classes— $\alpha$ -helical proteins and antiparallel  $\beta$  proteins—where no such predictive rules have been found. We will now examine a few examples that illustrate the relationship between the topology diagrams of some  $\alpha/\beta$  proteins, their switch points, and the active-site residues. These examples have been chosen because they represent different types of  $\alpha/\beta$  open-sheet structures.

### *Tyrosyl-tRNA synthetase has two different domains ( $\alpha/\beta + \alpha$ )*

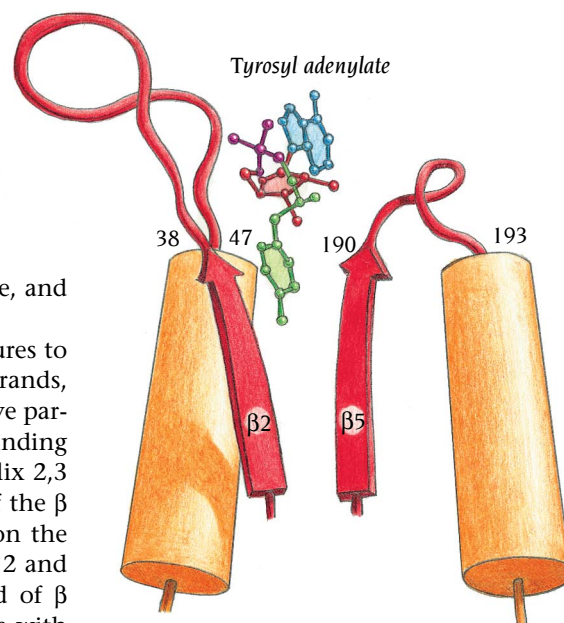
One of the crucial steps in protein synthesis is performed by the group of enzymes called aminoacyl-tRNA synthetases. These enzymes connect each amino acid with its specific transfer RNA molecule in a two-step reaction. First the amino acid is activated by ATP to give an enzyme-bound amino acid adenylate; then this complex is attacked by the tRNA to give the aminoacyl-tRNA.

The structure of the synthetase specific for the amino acid tyrosine was determined to 2.7 Å resolution in the laboratory of David Blow in London. Figure 4.15 shows a schematic diagram of the first 320 residues of a single subunit of this dimeric molecule. The last 100 residues are disordered in the crystal and are not visible. There are essentially two different domains, one



**Figure 4.15** Schematic diagram of the enzyme tyrosyl-tRNA synthetase, which couples tyrosine to its cognate transfer RNA. The central region of the catalytic domain (red and green) is an open twisted  $\alpha/\beta$  structure with five parallel  $\beta$  strands. The active site is formed by the loops from the carboxy ends of  $\beta$  strands 2 and 5. These two adjacent strands are connected to  $\alpha$  helices on opposite sides of the  $\beta$  sheet. Where more than one  $\alpha$  helix connects two  $\beta$  strands (for example, between strands 4 and 5), they are represented as one rectangle in the topology diagram. (Adapted from T.N. Bhat et al., *J. Mol. Biol.* 158: 699–709, 1982.)

**Figure 4.16** A schematic view of the active site of tyrosyl-tRNA synthetase. Tyrosyl adenylate, the product of the first reaction catalyzed by the enzyme, is bound to two loop regions: residues 38–47, which form the loop after  $\beta$  strand 2, and residues 190–193, which form the loop after  $\beta$  strand 5. The tyrosine and adenylate moieties are bound on opposite sides of the  $\beta$  sheet outside the carboxy ends of  $\beta$  strands 2 and 5.



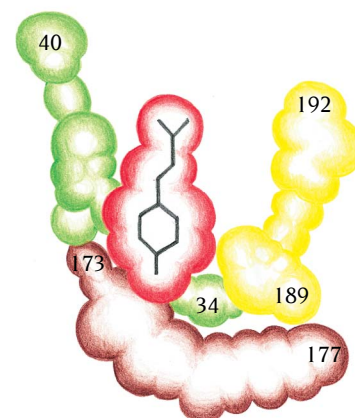
$\alpha/\beta$  domain (red and green in Figure 4.15) that binds ATP and tyrosine, and one  $\alpha$ -helical domain (blue), the function of which is not known.

Let us now apply the rules for predicting active sites of  $\alpha/\beta$  structures to the topological diagram shown in Figure 4.15. The  $\beta$  sheet has six strands, one of which, number 1, is antiparallel to the others. The remaining five parallel  $\beta$  strands are arranged in a way rather similar to the nucleotide-binding fold (see Figure 4.1b), but here the strand order is 6 5 2 3 4. Alpha helix 2,3 (which connects  $\beta$  strands 2 and 3) and  $\alpha$  helix 3,4 are on one side of the  $\beta$  sheet (red helices in Figure 4.15), whereas  $\alpha$  helices 4,5 and 5,6 are on the other side (green helices). The switch point is thus between  $\beta$  strands 2 and 5. We would predict that the active site is outside the carboxy end of  $\beta$  strands 2 and 5 and that the loop regions that connect these strands with their respective  $\alpha$  helices participate in binding the substrates. These loop regions comprise residues 38–47 and 190–193, respectively. The active site has been identified in the crystal structure by diffusing tyrosine and ATP into the crystals. The enzyme molecules in the crystals are active, so tyrosyl adenylate is formed, but because no tRNA is present, it stays bound to the enzyme.

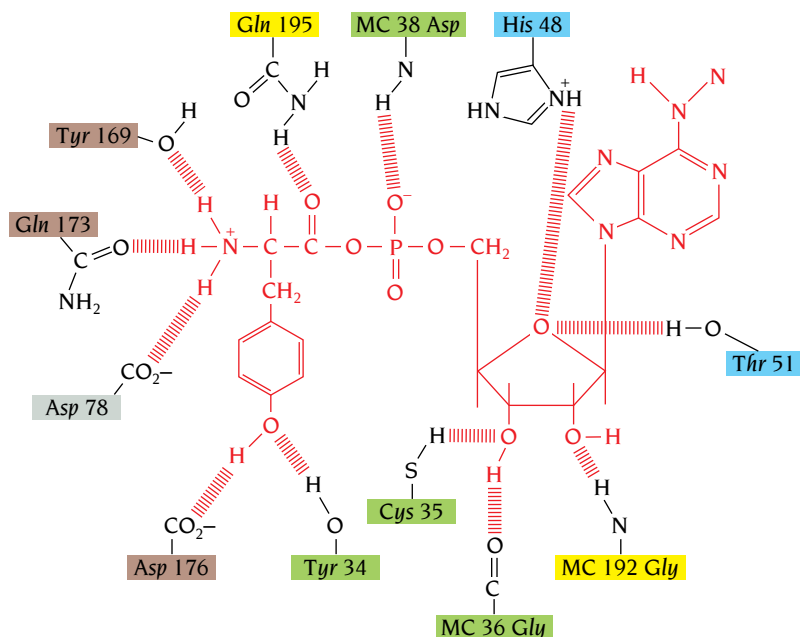
The position of this bound tyrosyl adenylate was determined from an electron density map of the complex just after the predictive rules were formulated. The region where it binds proved to be as predicted. Loop regions 38–47, after  $\beta$  strand 2, and 190–193, after  $\beta$  strand 5, line a cleft where the substrate binds (Figure 4.16). The phosphate and the sugar moieties are hydrogen-bonded to the main-chain nitrogen atoms of residues 38 and 192, respectively. This part of the substrate is thus very close to the switch point. The substrate straddles the edge of the  $\beta$  sheet so that the tyrosine and adenine ends are on opposite sides of the  $\beta$  sheet. The substrate also interacts with some of the other regions at this end of the  $\beta$  sheet, especially residues 173–177 of the  $\alpha$  helix that connects  $\beta$  strands 4 and 5 and, in addition, some residues within  $\beta$  strand 2. Figure 4.17 shows a schematic diagram of the position of bound tyrosine (red) in relation to these regions of the protein, and Figure 4.18 gives the important hydrogen bonds to the substrate, knowledge of which formed the basis for the beautiful site-directed mutagenesis experiments on this system by Alan Fersht in London.

### *Carboxypeptidase is an $\alpha/\beta$ protein with a mixed $\beta$ sheet*

Carboxypeptidases are zinc-containing enzymes that catalyze the hydrolysis of polypeptides at the C-terminal peptide bond. The bovine enzyme form A is a monomeric protein comprising 307 amino acid residues. The structure was determined in the laboratory of William Lipscomb, Harvard University, in 1970 and later refined to 1.5 Å resolution. Biochemical and x-ray studies have shown that the zinc atom is essential for catalysis by binding to the carbonyl oxygen of the substrate. This binding weakens the C=O bond by

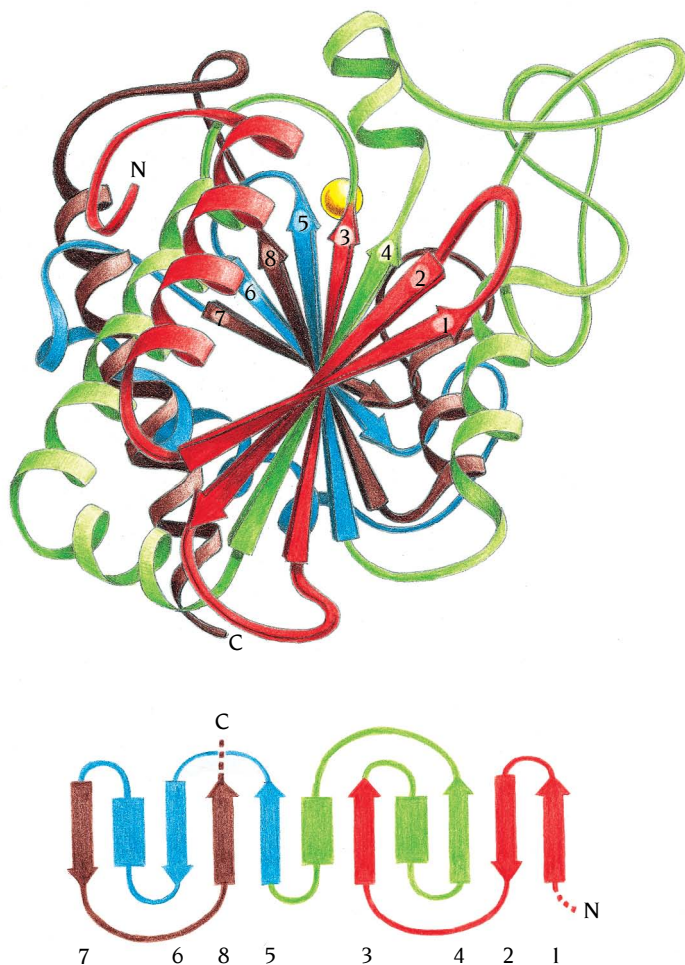


**Figure 4.17** Schematic diagram of bound tyrosine to tyrosyl-tRNA synthetase. Colored regions correspond to van der Waals radii of atoms within a layer of the structure through the tyrosine ring. Red is bound tyrosine; green is the end of  $\beta$  strand 2 and the beginning of the following loop region; yellow is the loop region 189–192; and brown is part of the  $\alpha$  helix in loop region 173–177.



**Figure 4.18** Side chains of the tyrosyl-tRNA synthetase that form hydrogen bonds to tyrosyl adenylate. Green residues are from  $\beta$  strand 2 and the following loop regions, yellow residues are from the loop after  $\beta$  strand 5, and brown residues are from the  $\alpha$  helix before  $\beta$  strand 5. (Adapted from T. Wells and A. Fersht, *Nature* 316: 656–657, 1985.)

abstracting electrons from the carbon atom and thus facilitates cleavage of the adjacent peptide bond. Carboxypeptidase is a large single domain structure comprising a mixed  $\beta$  sheet of eight  $\beta$  strands (Figure 4.19) with  $\alpha$  helices on both sides. Some of the loop regions are very long and curl around the central theme of the structure.



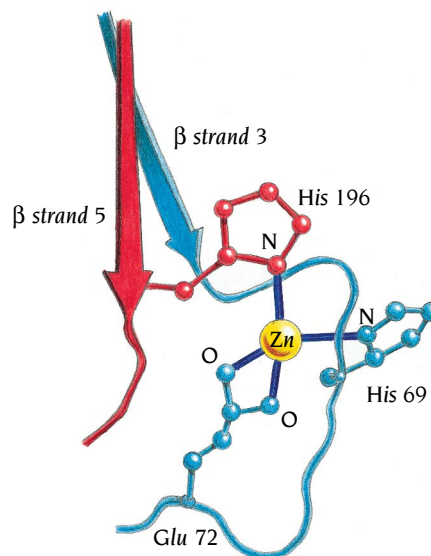
**Figure 4.19** Schematic and topological diagrams for the structure of the enzyme carboxypeptidase. The central region of the mixed  $\beta$  sheet contains four adjacent parallel  $\beta$  strands (numbers 8, 5, 3, and 4), where the strand order is reversed between strands 5 and 3. The active-site zinc atom (yellow circle) is bound to side chains in the loop regions outside the carboxy ends of these two  $\beta$  strands. The first part of the polypeptide chain is red, followed by green, blue, and brown. (Adapted from J. Richardson.)

**Figure 4.20** Detailed view of the zinc environment in carboxypeptidase. The active-site zinc atom is bound to His 69 and Glu 72, which are part of the loop region outside  $\beta$  strand 2. In addition, His 196, which is the last residue of  $\beta$  strand 5, also binds the zinc.

The four central strands of the  $\beta$  sheet are parallel and have the strand order 8 5 3 4 (see Figure 4.19). The strand order is thus reversed once, and there is a switch point in the middle of this  $\beta$  sheet between  $\beta$  strands 5 and 3 where we would expect the active site to be located.

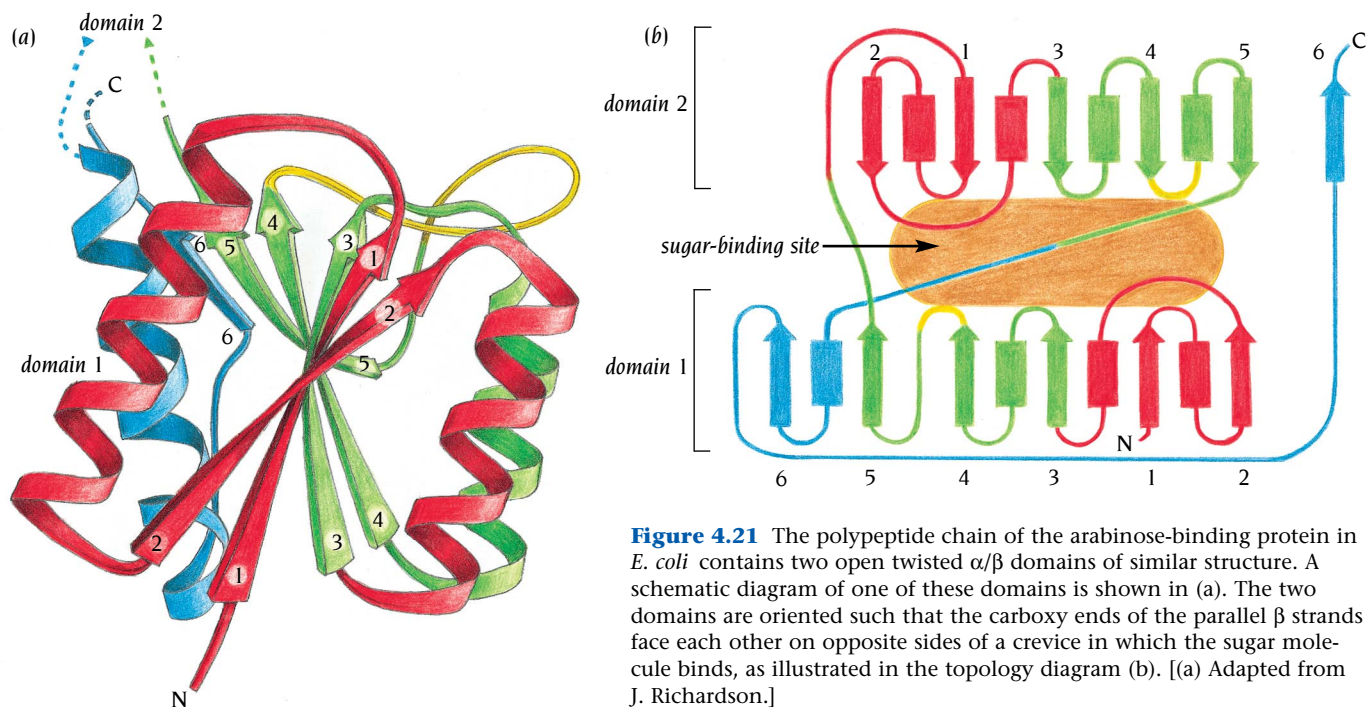
This is precisely where the catalytically essential zinc atom is found. This zinc atom is located precisely at this switch point, where it is firmly anchored to the protein by three side-chain ligands, His 69, Glu 72, and His 196 (Figure 4.20). The last residue of  $\beta$  strand 3 is residue 66, so the two zinc ligands His 69 and Glu 72 are at the beginning of the loop region that connects this  $\beta$  strand with its corresponding  $\alpha$  helix. The last residue of  $\beta$  strand 5 is the third zinc ligand, His 196.

In this structure the loop regions adjacent to the switch point do not provide a binding crevice for the substrate but instead accommodate the active-site zinc atom. The essential point here is that this zinc atom and the active site are in the predicted position outside the switch point for the four central parallel  $\beta$  strands, even though these  $\beta$  strands are only a small part of the total structure. This sort of arrangement, in which an active site formed from parallel  $\beta$  strands is flanked by antiparallel  $\beta$  strands, has been found in a number of other  $\alpha/\beta$  proteins with mixed  $\beta$  sheets.

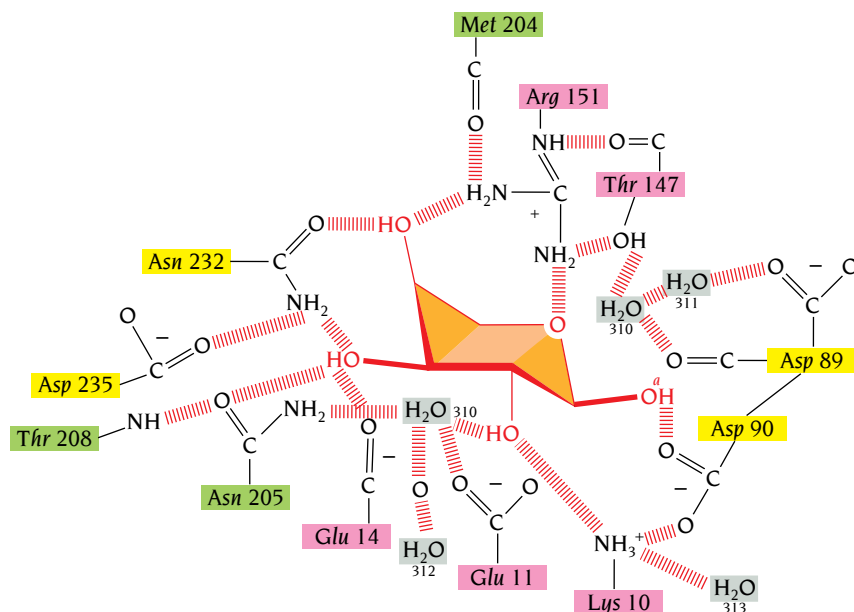


### Arabinose-binding protein has two similar $\alpha/\beta$ domains

The arabinose-binding protein is one of the group of proteins that occur in the periplasmic space between the inner and outer cell membranes of Gram-negative bacteria such as *E. coli*. These proteins are components of active transport systems for various sugars, amino acids, and ions. Arabinose-binding protein is involved in arabinose transport. It is a single polypeptide chain of 306 amino acids folded into two domains of similar structure and topology (Figure 4.21), as was ascertained in the laboratory of Florante Quiocho in Houston, Texas, by a structure determination of the protein with bound



**Figure 4.21** The polypeptide chain of the arabinose-binding protein in *E. coli* contains two open twisted  $\alpha/\beta$  domains of similar structure. A schematic diagram of one of these domains is shown in (a). The two domains are oriented such that the carboxy ends of the parallel  $\beta$  strands face each other on opposite sides of a crevice in which the sugar molecule binds, as illustrated in the topology diagram (b). [(a) Adapted from J. Richardson.]



**Figure 4.22** Schematic diagram of the complex networks of hydrogen bonds formed by polar side chains from the arabinose-binding protein and L-arabinose. The residues that interact with the sugar are in turn hydrogen-bonded to each other or to other residues or isolated water molecules. The pink and green residues are in loop regions that, from the topology diagram, are predicted to form the binding site. The yellow residues are from adjacent loop regions. (Adapted from C.F. Sams et al., *Nature* 310: 429–430, 1984.)

arabinose at 1.7 Å resolution. The first five  $\beta$  strands in both domains are parallel; the strand order is reversed once, and the switch points are between  $\beta$  strands 1 and 3 in both domains.

A schematic diagram relating binding to the topology of the domains is shown in Figure 4.21b. Twelve amino acid residues from both domains are involved in forming a complicated network of hydrogen bonds to the oxygen atoms of the bound arabinose (Figure 4.22). We predict from the topology of these domains that loop residues from  $\beta$  strands 1 and 3 of both domains should participate in these interactions. Five residues in loop regions after  $\beta$  strand 1 in both domains (pink residues in Figure 4.22) and three residues after  $\beta$  strand 3 in the second domain (green residues in Figure 4.22) participate in binding. The remaining four residues (yellow residues in Figure 4.22) are all from loop regions after  $\beta$  strand 4 in both domains. These are one  $\beta$  strand removed from the switch point. Both domains thus participate approximately equally in binding the sugar at the carboxy edge of the  $\beta$  sheets, outside the switch point, where the strand order is reversed.

A number of proteins consist of two  $\alpha/\beta$  open-sheet domains formed from a single polypeptide chain, as in arabinose-binding protein. In almost all these cases the active sites are found in cleft regions between these two domains. The domains are oriented in such a way that the carboxy edge of both  $\beta$  sheets points toward the active site. Loop regions adjacent to the switch points of both domains participate in forming the active site. In enzymatic reactions where two different substrates participate, they are bound to different domains and are brought together for catalytic reactions by the orientation of these domains. In other proteins the two domains bind different regions of the same ligand. Sugar-binding proteins from bacteria are examples of this second case.

## Conclusion

Alpha/beta ( $\alpha/\beta$ ) structures are the most frequent and most regular of the protein structures. They fall into three classes: the first class comprises a central core of usually eight parallel  $\beta$  strands arranged close together like the staves of a barrel, surrounded by  $\alpha$  helices; the second class comprises an open twisted parallel or mixed  $\beta$  sheet with  $\alpha$  helices on both sides of the  $\beta$  sheet; and the third class is formed by leucine-rich motifs in which a large number of parallel  $\beta$  strands form a curved  $\beta$  sheet with all the  $\alpha$  helices on the outside of this sheet.

The  $\alpha/\beta$ -barrel structure is one of the largest and most regular of all domain structures, comprising about 250 amino acids. It has so far been found in more than 20 different proteins, with completely different amino acid sequences and different functions. They are all enzymes that are modeled on this common scaffold of eight parallel  $\beta$  strands surrounded by eight  $\alpha$  helices. They all have their active sites in very similar positions, at the bottom of a funnel-shaped pocket created by the loops that connect the carboxy end of the  $\beta$  strands with the amino end of the  $\alpha$  helices. The specific enzymatic activity is, in each case, determined by the lengths and amino acid sequences of these loop regions, which do not contribute to the stability of the fold.

The horseshoe structure is formed by homologous repeats of leucine-rich motifs, each of which forms a  $\beta$ -loop- $\alpha$  unit. The units are linked together such that the  $\beta$  strands form an open curved  $\beta$  sheet, like a horseshoe, with the  $\alpha$  helices on the outside of the  $\beta$  sheet and the inside exposed to solvent. The invariant leucine residues of these motifs form the major part of the hydrophobic region between the  $\alpha$  helices and the  $\beta$  sheet.

The open  $\alpha/\beta$ -sheet structures vary considerably in size, number of  $\beta$  strands, and strand order. Independent of these variations, they all have their active sites at the carboxy edge of the  $\beta$  strands, and these active sites are lined by the loop regions that connect the  $\beta$  strands with the  $\alpha$  helices. In this respect, they are similar to the  $\alpha/\beta$ -barrel structures. However, the active-site regions are created differently in open structures. They are formed in those regions outside the carboxy edge of the  $\beta$  sheet, where two adjacent loops are on opposite sides of the  $\beta$  sheet. The positions of these regions can be predicted from topology diagrams. The rules that relate the general position of functional binding sites to the overall structure of the protein are thus known for  $\beta$  barrels and open-sheet  $\alpha/\beta$  proteins.

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### Specific structures

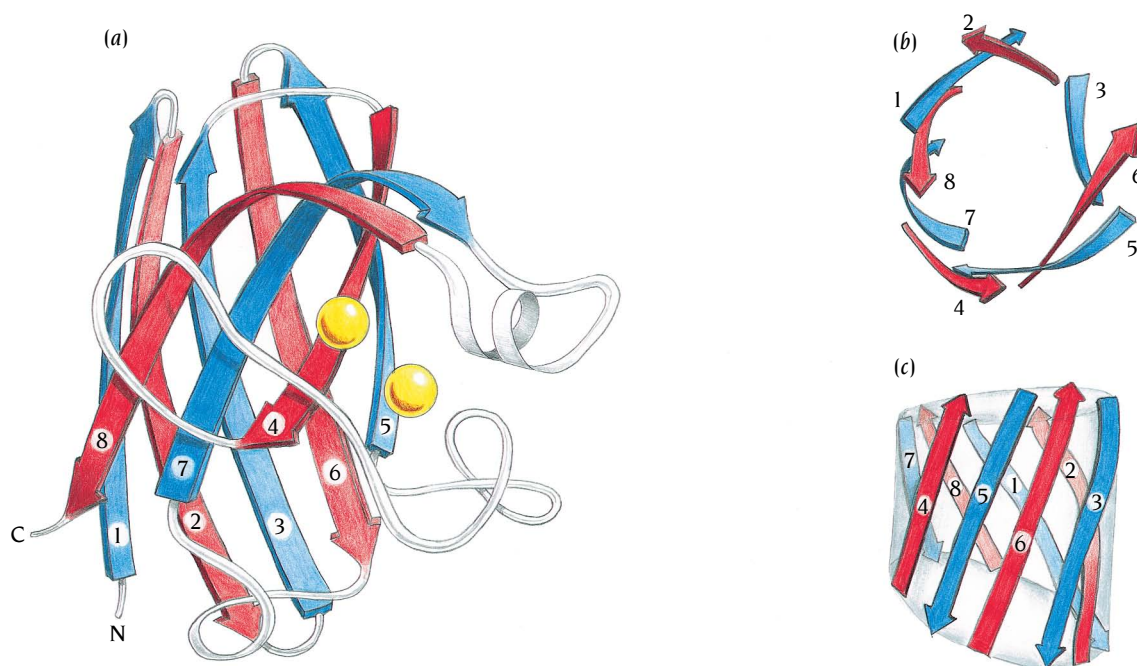
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Antiparallel beta ( $\beta$ ) structures comprise the second large group of protein domain structures. Functionally, this group is the most diverse; it includes enzymes, transport proteins, antibodies, cell surface proteins, and virus coat proteins. The cores of these domains are built up by  $\beta$  strands that can vary in number from four or five to over ten. The  $\beta$  strands are arranged in a predominantly antiparallel fashion and usually in such a way that they form two  $\beta$  sheets that are joined together and packed against each other.

The  $\beta$  sheets have the usual twist, and when two such twisted  $\beta$  sheets are packed together, they form a barrel-like structure (Figure 5.1). Antiparallel  $\beta$  structures, therefore, in general have a core of hydrophobic side chains inside the barrel provided by residues in the  $\beta$  strands. The surface is formed by residues from the loop regions and from the strands. The aim of this chapter is to examine a number of antiparallel  $\beta$  structures and demonstrate how these rather complex structures can be separated into smaller comprehensible motifs.

**Figure 5.1** The enzyme superoxide dismutase (SOD). SOD is a  $\beta$  structure comprising eight antiparallel  $\beta$  strands (a). In addition, SOD has two metal atoms, Cu and Zn (yellow circles), that participate in the catalytic action: conversion of a superoxide radical to hydrogen peroxide and oxygen. The eight  $\beta$  strands are arranged around the surface of a barrel, which is viewed along the barrel axis in (b) and perpendicular to this axis in (c). [(a) Adapted from J.S. Richardson. The structure of SOD was determined in the laboratory of J.S. and D.R. Richardson, Duke University.]



In Chapter 2 we described the 24 different ways that two  $\beta$ -loop- $\beta$  units can form a four-stranded  $\beta$  sheet. The number of possible ways to form antiparallel  $\beta$ -sheet structures rapidly increases as the number of strands increases. It is thus surprising, but reassuring, that the number of topologies actually observed is small and that most  $\beta$  structures fall into a few groups of common or similar topology. The three most frequently occurring groups—up-and-down barrels, Greek keys, and jelly roll barrels—can all be related to simple ways of connecting antiparallel  $\beta$  strands arranged in a barrel structure.

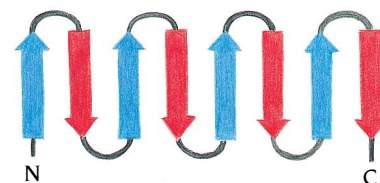
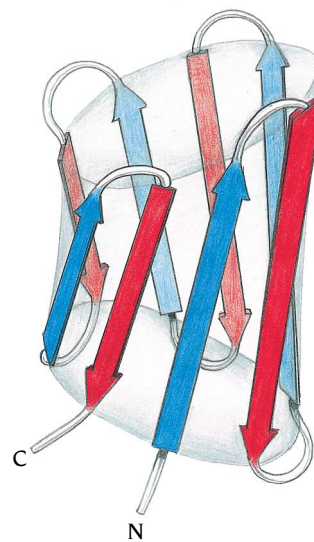
### *Up-and-down barrels have a simple topology*

The simplest topology is obtained if each successive  $\beta$  strand is added adjacent to the previous strand until the last strand is joined by hydrogen bonds to the first strand and the barrel is closed (Figure 5.2). These are called **up-and-down  $\beta$  sheets** or **barrels**. The arrangement of  $\beta$  strands is similar to that in the  $\alpha/\beta$ -barrel structures we have just described in Chapter 4, except that here the strands are antiparallel and all the connections are hairpins. The structural and functional versatility of even this simple arrangement will be illustrated by two examples.

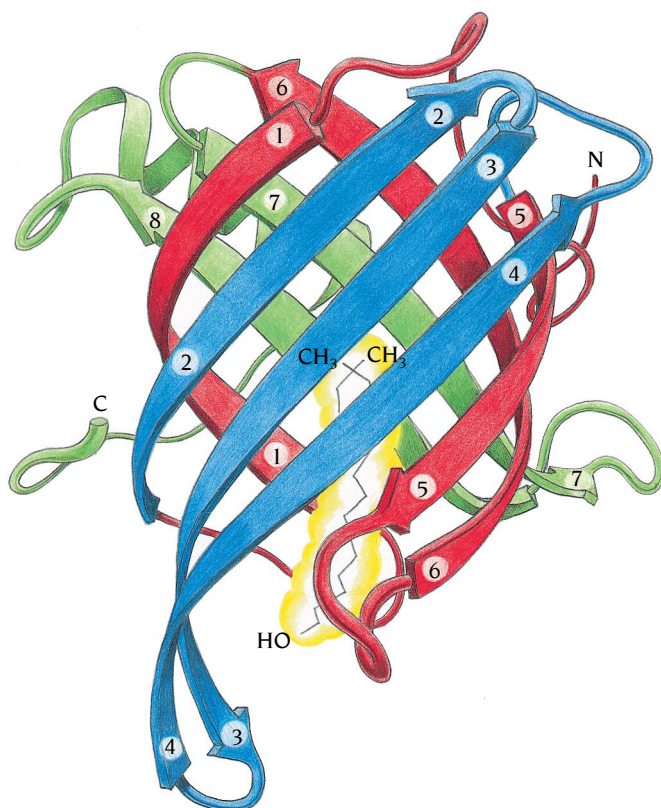
### *The retinol-binding protein binds retinol inside an up-and-down $\beta$ barrel*

The first example is the plasma-borne **retinol-binding protein**, RBP, which is a single polypeptide chain of 182 amino acid residues. This protein is responsible for transporting the lipid alcohol **vitamin A** (retinol) from its storage site in the liver to the various vitamin-A-dependent tissues. It is a disposable package in the sense that each RBP molecule transports only a single retinol molecule and is then degraded.

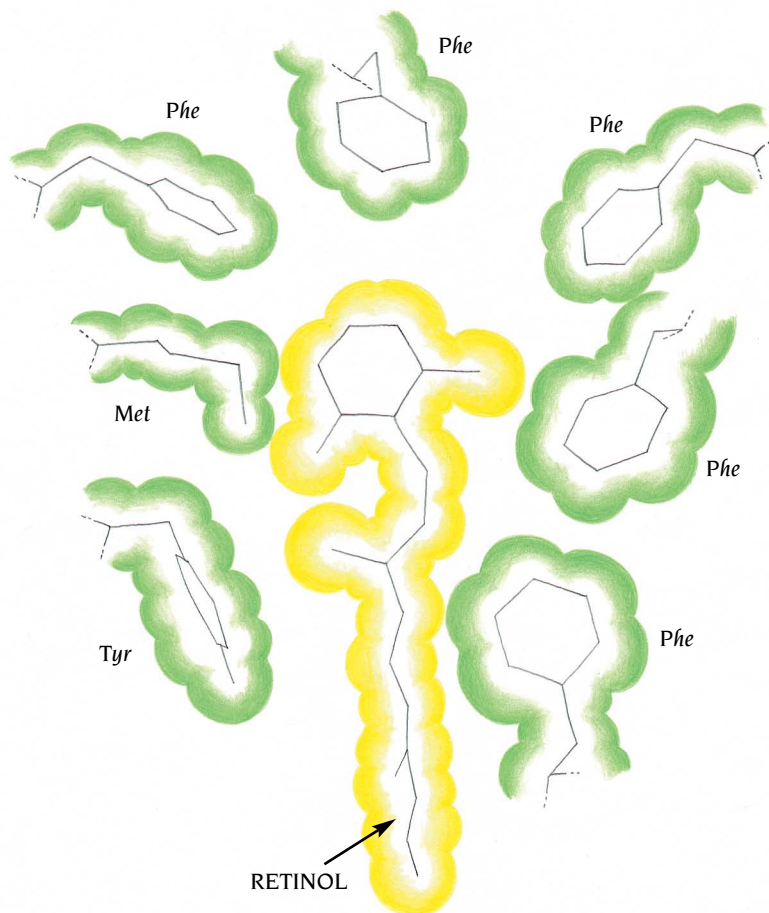
RBP is synthesized in the hepatocytes, where it picks up one molecule of retinol in the endoplasmic reticulum. Both its synthesis and its secretion from the hepatocytes to the plasma are regulated by retinol. In plasma, the



**Figure 5.2** Schematic and topological diagrams of an up-and-down  $\beta$  barrel. The eight  $\beta$  strands are all antiparallel to each other and are connected by hairpin loops. Beta strands that are adjacent in the amino acid sequence are also adjacent in the three-dimensional structure of up-and-down barrels.



**Figure 5.3** Schematic diagram of the structure of human plasma retinol-binding protein (RBP), which is an up-and-down  $\beta$  barrel. The eight antiparallel  $\beta$  strands twist and curl such that the structure can also be regarded as two  $\beta$  sheets (green and blue) packed against each other. Some of the twisted  $\beta$  strands (red) participate in both  $\beta$  sheets. A retinol molecule, vitamin A (yellow), is bound inside the barrel, between the two  $\beta$  sheets, such that its only hydrophilic part (an OH tail) is at the surface of the molecule. The topological diagram of this structure is the same as that in Figure 5.2. (Courtesy of Alwyn Jones, Uppsala, Sweden.)



**Figure 5.4** The binding site for retinol inside the RBP barrel is lined with hydrophobic residues. They provide a hydrophobic surrounding for the hydrophobic part of the retinol molecule.

RBP-retinol complex binds to a larger protein molecule, prealbumin, which further stabilizes it and prevents its loss via the kidney. Recognition of this complex by a cell-surface receptor causes RBP to release the retinol and, as a result, to undergo a conformational change that drastically reduces its affinity for prealbumin. The free RBP molecule is then excreted through the kidney glomerus, reabsorbed in the proximal tubule cells, and degraded.

The structure of RBP with bound retinol has been determined in the laboratory of Alwyn Jones in Uppsala, Sweden to 2.0 Å resolution. Its most striking feature is a  $\beta$ -barrel core consisting of eight up-and-down antiparallel  $\beta$  strands as shown in Figure 5.3. In addition, there is a four-turn  $\alpha$  helix at the carboxy end of the polypeptide chain that is packed against the outside of the  $\beta$  barrel. The  $\beta$  strands are curved and twisted, and the barrel is wrapped around the retinol molecule. One end of the barrel is open to the solvent whereas the other end is closed by tight side-chain packing: the tail of the retinol molecule is at the open end of the barrel.

The hydrophobic retinol molecule is packed against hydrophobic side chains from the  $\beta$  strands in the barrel's core (Figure 5.4). The structure of the apo-form where the retinol molecule is removed is also known. Surprisingly, there is almost no change in the barrel structure, and a large hole is left inside the barrel.

### *Amino acid sequence reflects $\beta$ structure*

On a large part of the surface of RBP (the front face in Figure 5.3), side chains from residues in the  $\beta$  strands are exposed to the solvent. This is achieved by alternating hydrophobic with polar or charged hydrophilic residues in the

| strand no. | residue no. | amino acid sequence                                                               |     |                                                                                   |     |                                                                                   |     |                                                                                   |     |   |     |   |     |   |     |   |     |   |
|------------|-------------|-----------------------------------------------------------------------------------|-----|-----------------------------------------------------------------------------------|-----|-----------------------------------------------------------------------------------|-----|-----------------------------------------------------------------------------------|-----|---|-----|---|-----|---|-----|---|-----|---|
|            |             |  |     |  |     |  |     |  |     |   |     |   |     |   |     |   |     |   |
| 2          | 41–48       | –                                                                                 | Ile | –                                                                                 | Val | –                                                                                 | Ala | –                                                                                 | Glu | – | Phe | – | Ser | – | Val | – | Asp | – |
| 3          | 53–60       | –                                                                                 | Met | –                                                                                 | Ser | –                                                                                 | Ala | –                                                                                 | Thr | – | Ala | – | Lys | – | Gly | – | Arg | – |
| 4          | 71–78       | –                                                                                 | Ala | –                                                                                 | Asp | –                                                                                 | Met | –                                                                                 | Val | – | Gly | – | Thr | – | Phe | – | Thr | – |

**Figure 5.5** Amino acid sequence of  $\beta$  strands 2 3 4 in human plasma retinol-binding protein. The sequences are listed in such a way that residues which point into the barrel are arrowed and colored green. The remaining residues are exposed to the solvent.

amino acid sequences of the  $\beta$  strands; in other words, side chains of the  $\beta$  strands form the hydrophobic core of the barrel as well as part of the hydrophilic outer surface. Strands 2 3 4 of RBP clearly illustrate this arrangement (Figure 5.5, where the core residues are colored green). This structure also very clearly illustrates that an antiparallel  $\beta$  barrel is built up from two  $\beta$  sheets that are packed against each other (see Figure 5.3). Beta strands 1 2 3 4 5 6 (blue and red color) form one sheet, and strands 1 8 7 6 5 (green and red color) form the second sheet. Strands 1 5 6 thus contribute to both sheets by having sharp corners where they can turn over from one sheet to the other.

### *The retinol-binding protein belongs to a superfamily of protein structures*

RBP is one member of a **superfamily** of proteins with different functions, marginally homologous amino acid sequences, but similar three-dimensional structures. This superfamily also includes an insect protein that binds a blue pigment, biliverdin, and  $\beta$ -lactoglobulin, a protein that is abundant in milk. All have polypeptide chains of approximately the same lengths that are wrapped into very similar up-and-down eight-stranded antiparallel  $\beta$  barrels. They all tightly bind hydrophobic ligands inside this barrel.

There is a second family of small lipid-binding proteins, the **P2 family**, which include among others cellular retinol- and fatty acid-binding proteins as well as a protein, P2, from myelin in the peripheral nervous system. However, members of this second family have ten antiparallel  $\beta$  strands in their barrels compared with the eight strands found in the barrels of the RBP superfamily. Members of the P2 family show no amino acid sequence homology to members of the RBP superfamily. Nevertheless, their three-dimensional structures have similar architecture and topology, being up-and-down  $\beta$  barrels.

### *Neuraminidase folds into up-and-down $\beta$ sheets*

A second example of up-and-down  $\beta$  sheets is the protein neuraminidase from influenza virus. Here the packing of the sheets is different from that in RBP. They do not form a simple barrel but instead six small sheets, each with four  $\beta$  strands, which are arranged like the blades of a six-bladed propeller. Loop regions between the  $\beta$  strands form the active site in the middle of one side of the propeller. Other similar structures are known with different numbers of the same motif arranged like propellers with different numbers of blades such as the G-proteins discussed in Chapter 13.

Influenza virus is an RNA virus with an outer lipid envelope. There are two viral proteins anchored in this membrane, neuraminidase and hemagglutinin. They are both transmembrane proteins with a few residues inside the membrane and a transmembrane region followed by a stalk and a head-piece outside the membrane. The heads are exposed on the surface of the virion and thus provide the antigenic determinants of this epidemic virus. The function of hemagglutinin, which is glycosylated, is to mediate the binding of virus particles to host cells by recognizing and binding to sialic acid residues on glycoproteins of the cell membrane, as we shall discuss in more detail at the end of this chapter.

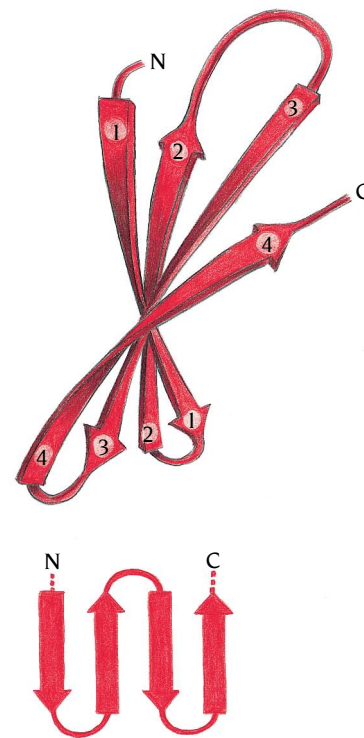
The role of the viral neuraminidase, conversely, seems to be to facilitate the release of progeny virions from infected cells by cleaving sialic acid

residues from the carbohydrate side chains both of the viral hemagglutinin and of the glycosylated cellular membrane proteins. This helps prevent progeny virions from binding to and reinfecting the cells from whose surface they have just budded. From the point of view of the viruses, reinfecting an already infected cell is, of course, a waste of time.

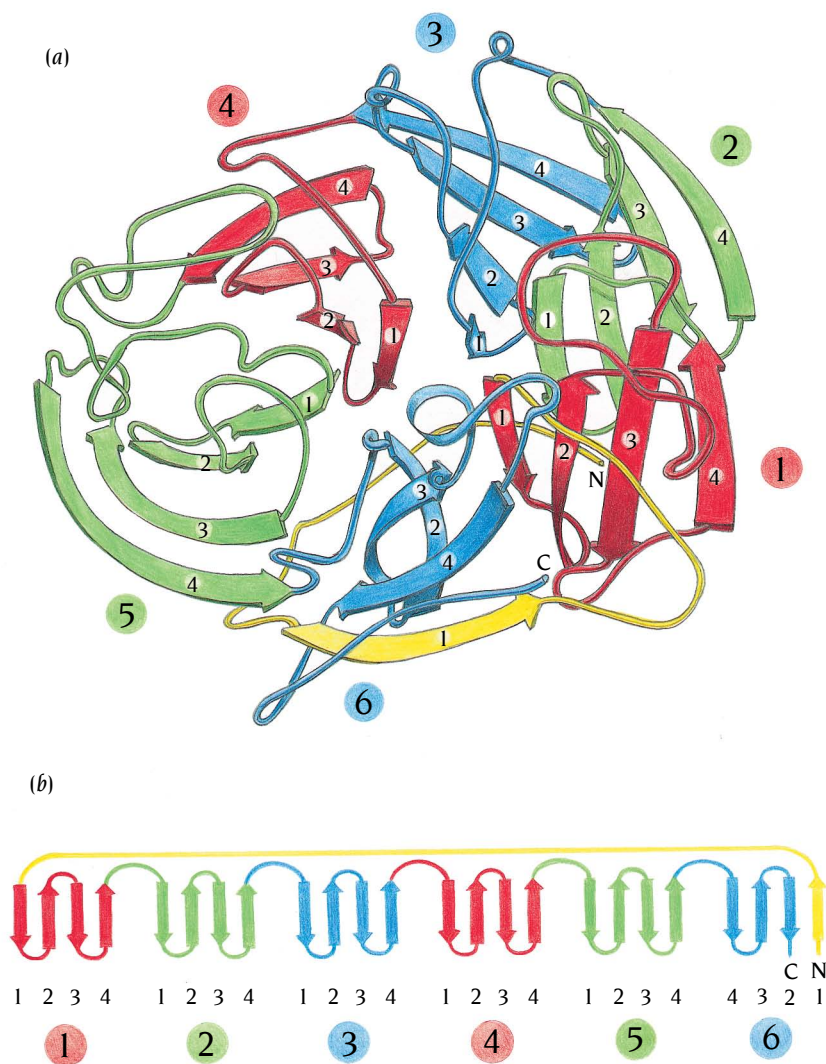
The neuraminidase molecule is a homotetramer made up of four identical polypeptide chains, each of around 470 amino acids; the exact number varies depending on the strain of the virus. If influenza virus is treated with the proteolytic enzyme pronase, the head of the neuraminidase, which is soluble, is cleaved off from the stalk projecting from the viral envelope. The soluble head, comprising four subunits of about 400 amino acids each, can be crystallized.

### *Folding motifs form a propeller-like structure in neuraminidase*

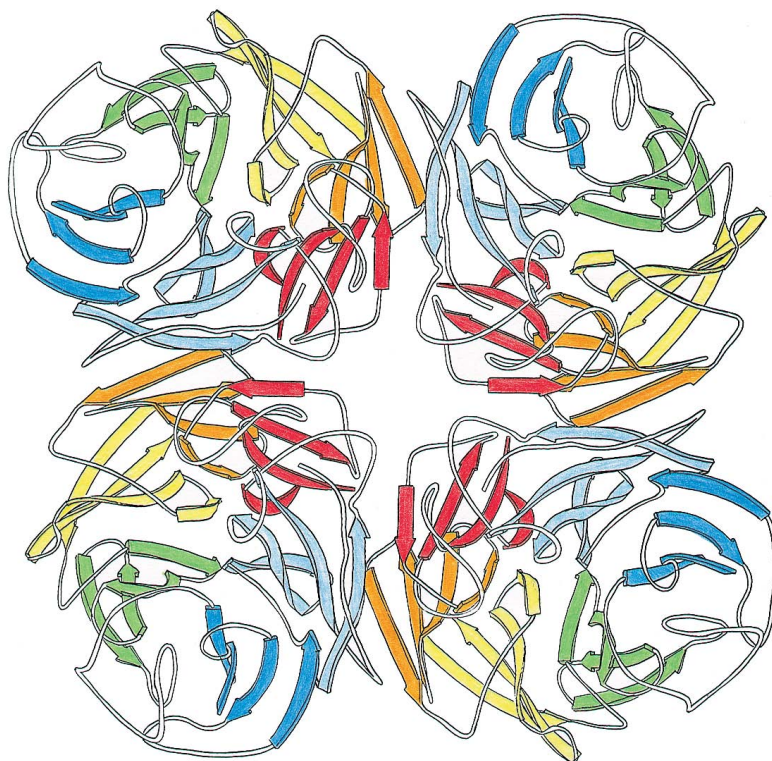
The structure of these tetrameric neuraminidase heads was determined in the laboratory of Peter Colman in Parkville, Australia to 2.9 Å resolution. Each of the four subunits of the tetramer is folded into a single domain built up from six closely packed, similarly folded motifs. The motif is a simple up-and-down antiparallel  $\beta$  sheet of four strands (Figure 5.6). The strands have a rather large twist such that the directions of the first and the fourth strands differ by 90°. To a first very rough approximation the six motifs are arranged within each subunit with an approximate sixfold symmetry around an axis through the center of the subunit (Figure 5.7a). These six  $\beta$  sheets are arranged like six blades of a propeller.



**Figure 5.6** Schematic and topological diagrams of the folding motif in neuraminidase from influenza virus. The motif is built up from four antiparallel  $\beta$  strands joined by hairpin loops, an up-and-down open  $\beta$  sheet.



**Figure 5.7** The subunit structure of the neuraminidase headpiece (residues 84–469) from influenza virus is built up from six similar, consecutive motifs of four up-and-down antiparallel  $\beta$  strands (Figure 5.6). Each such motif has been called a propeller blade and the whole subunit structure a six-blade propeller. The motifs are connected by loop regions from  $\beta$  strand 4 in one motif to  $\beta$  strand 1 in the next motif. The schematic diagram (a) is viewed down an approximate sixfold axis that relates the centers of the motifs. Four such six-blade propeller subunits are present in each complete neuraminidase molecule (see Figure 5.8). In the topological diagram (b) the yellow loop that connects the N-terminal  $\beta$  strand to the first  $\beta$  strand of motif 1 is not to scale. In the folded structure it is about the same length as the other loops that connect the motifs. (Adapted from J. Varghese et al., *Nature* 303: 35–40, 1983.)



**Figure 5.8** Schematic view down the fourfold axis of the tetrameric molecule of neuraminidase as it appeared on the cover of *Nature*, May 5, 1983.

In summary, the whole molecule has almost 1600 amino acid residues. It is composed of four identical polypeptide chains, each of which is folded into a superbarrel with 24  $\beta$  strands (Figure 5.8). These 24  $\beta$  strands are arranged in six similar motifs, each of which contains four  $\beta$  strands that form the blades of a propeller-like structure.

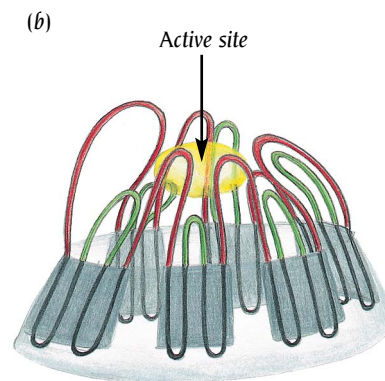
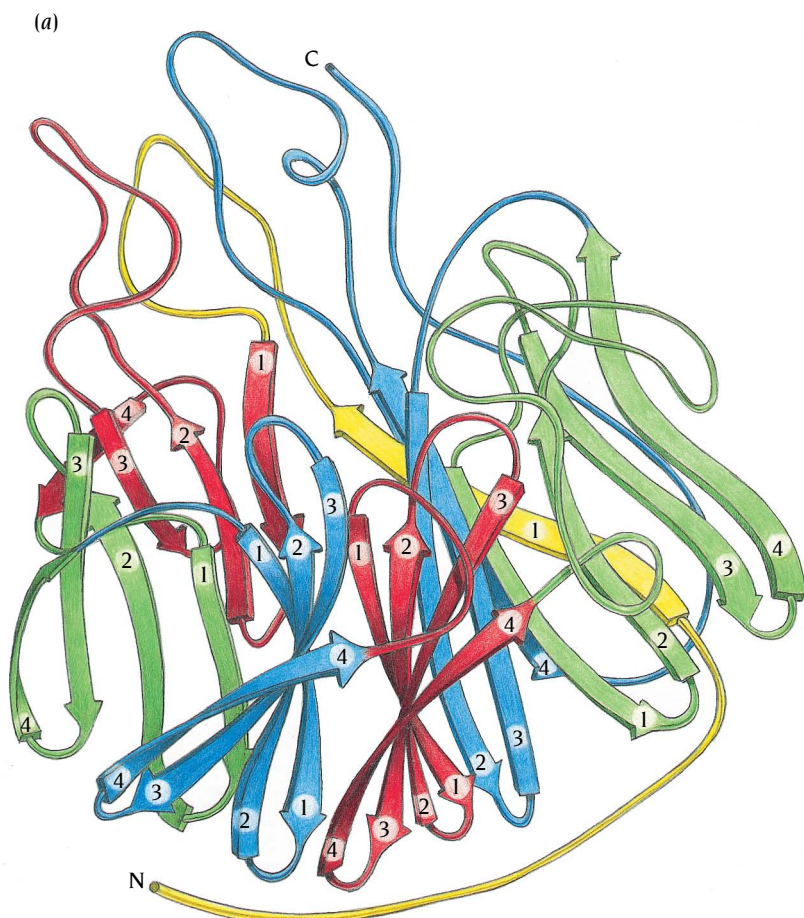
### *The active site is in the middle of one side of the propeller*

Not only are the topologies within the six  $\beta$  sheets in each subunit identical, but so are their connections to each other, with the exception of the last  $\beta$  sheet (see Figure 5.7b). The fourth strand of each  $\beta$  sheet is connected across the top of the subunit (seen in Figure 5.7a coming out of the page) to the first strand of the next sheet. The loop that connects strands 2 and 3 within the sheet is also at the top of the subunit.

Furthermore, because of the approximate sixfold symmetry of the  $\beta$ -sheet motifs, these 12-loop regions, derived from the six  $\beta$  sheets, are on the same side of the molecule, as can be seen in Figure 5.9a, where we see a single polypeptide chain (one of the four subunits) from the side of the propeller. The  $\beta$  sheets are arranged cyclically around an axis through the center of the molecule. The loop regions at the top of this barrel are extensive (Figure 5.9a) and together they form a wide funnel-shaped pocket containing the active site (Figure 5.9b). This is analogous to the active site formed by the loop regions at the top of the  $\alpha/\beta$ -barrel structures.

### *Greek key motifs occur frequently in antiparallel $\beta$ structures*

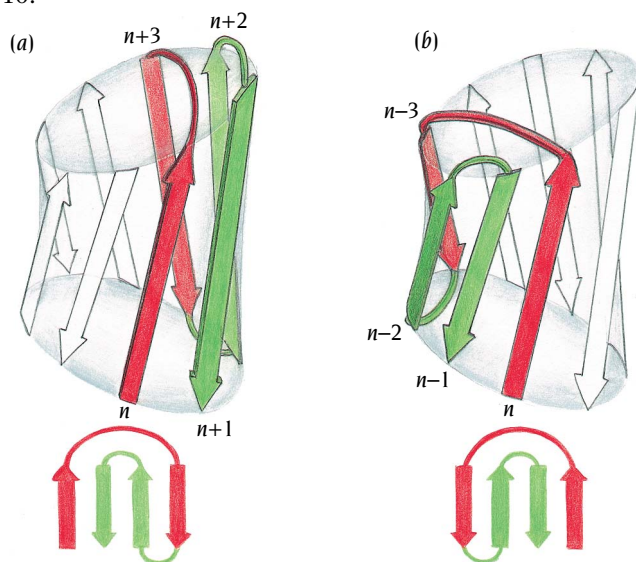
We saw in Chapter 2 that the **Greek key motif** provides a simple way to connect antiparallel  $\beta$  strands that are on opposite sides of a barrel structure. We will now look at how this motif is incorporated into some of the simple antiparallel  $\beta$ -barrel structures and show that an antiparallel  $\beta$  sheet of eight strands can be built up only by hairpin and/or Greek key motifs, if the connections do not cross between the two ends of the  $\beta$  sheet.



**Figure 5.9** The six four-stranded motifs in a single subunit of neuraminidase form the six blades of a propeller-like structure. A schematic diagram of the subunit structure shows the propeller viewed from its side (a). An idealized propeller structure viewed from the side to highlight the position of the active site is shown in (b). The loop regions that connect the motifs (red in b) in combination with the loops that connect strands 2 and 3 within the motifs (green in b) form a wide funnel-shaped active site pocket. [(a) Adapted from P. Colman et al., *Nature* 326: 358–363, 1987.]

Assume that we have eight antiparallel  $\beta$  strands arranged in a barrel structure. We decide that we want to connect strand number  $n$  to an antiparallel strand at the same end of the barrel. We do not want to connect it to strand number  $n + 1$  as in the up-and-down barrels just described, nor do we want to connect it to strand number  $n - 1$  which is equivalent to turning the up-and-down barrel in Figure 5.2 upside down. What alternatives remain?

It is easy to see from Figure 5.10 that there are only two alternatives. We can connect it either to strand number  $n + 3$  or to  $n - 3$ . Both cases require only short loop regions that traverse the end of the barrel. How do we now continue the connections? The simplest way to connect the strands that were skipped over is to join them by up-and-down connections, as illustrated in Figure 5.10.



**Figure 5.10** Idealized diagrams of the Greek key motif. This motif is formed when one of the connections of four antiparallel  $\beta$  strands is not a hairpin connection. The motif occurs when strand number  $n$  is connected to strand  $n + 3$  (a) or  $n - 3$  (b) instead of  $n + 1$  or  $n - 1$  in an eight-stranded antiparallel  $\beta$  sheet or barrel. The two different possible connections give two different hands of the Greek key motif. In all protein structures known so far only the hand shown in (a) has been observed.

We have now connected four adjacent strands of the barrel in a simple and logical fashion requiring only short loop regions. The result is the Greek key motif described in Chapter 2, which is found in the large majority of antiparallel  $\beta$  structures. The two cases represent the two possible different hands, but in all structures known to us the hand that corresponds to the case where  $\beta$  strand  $n$  is linked to  $\beta$  strand  $n + 3$  as in Figure 5.10a is present.

The remaining four strands of the barrel can be joined either by up-and-down connections before and after the motif or by another Greek key motif. We will examine examples of both cases.

### The $\gamma$ -crystallin molecule has two domains

The transparency and refractive power of the lenses of our eyes depend on a smooth gradient of refractive index for visible light. This is achieved partly by a regular packing arrangement of the cells in the lens and partly by a smoothly changing concentration gradient of lens-specific proteins, the crystallins.

There are at least three different classes of crystallins. The  $\alpha$  and  $\beta$  are heterogeneous assemblies of different subunits specified by different genes, whereas the **gamma ( $\gamma$ ) crystallins** are monomeric proteins with a polypeptide chain of around 170 amino acid residues. The structure of one such  $\gamma$  crystallin was determined in the laboratory of Tom Blundell in London to 1.9 Å resolution. A picture of this molecule generated from a graphics display is shown in Figure 5.11.

Let us now examine this molecule and dissect it into its structural components to see if we can understand how these are put together. We will reduce this rather complex, and at first sight bewildering, structure to its simplest representation as a series of motifs. This will help us to understand the structure and see its relationships to other structures.

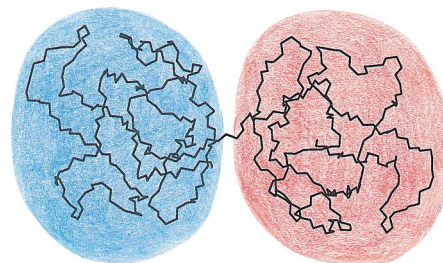
We can immediately discern from Figure 5.11 that the molecule is divided into two clearly separated domains that seem to be of similar size. For the next step we would need a stereopicture of the model or, much better, a graphics display where we could manipulate the model and look at it from different viewpoints. Here instead we have made a schematic diagram of one domain (Figure 5.12), which is normally not done until the analysis is completed and the structural principles are clear.

### The domain structure has a simple topology

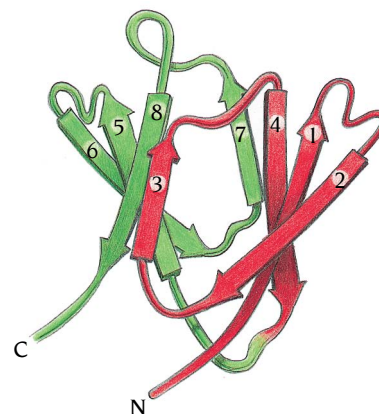
We will now follow the main polypeptide chain and trace out a topological diagram for this domain. We can immediately see from Figure 5.13 that the only secondary structure in the molecule is made up of  $\beta$  strands, which are arranged in an antiparallel fashion into two separate  $\beta$  sheets. Beta strands 1, 2, 4, and 7 form one antiparallel  $\beta$  sheet with the strand order 2 1 4 7. We thus draw the left four arrows in Figure 5.13 and connect strands 1 and 2. Similarly, we see that  $\beta$  strands 3, 5, 6, and 8 form another antiparallel  $\beta$  sheet with the strand order 6 5 8 3. We notice that strands 7 and 6 are adjacent although not hydrogen bonded to each other on the back side of the domain. We thus position strand 6 adjacent to strand 7 in the topology diagram but make a space between them to indicate that they belong to different  $\beta$  sheets. Alternatively, we could have positioned strands 2 and 3 adjacent to each other, which would have given a topologically identical diagram. We then connect the strands in consecutive order along the polypeptide chain.

### Two Greek key motifs form the domain

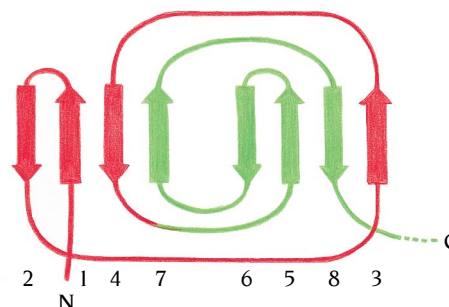
The topological diagram of Figure 5.13 has been drawn to reflect the observation that the two  $\beta$  sheets are separate:  $\beta$  strands 2 and 3 are not hydrogen bonded to each other, nor are strands 6 and 7. The connections



**Figure 5.11** A computer-generated diagram of the structure of  $\gamma$ -crystallin comprising one polypeptide chain of 170 amino acid residues. The diagram illustrates that the polypeptide chain is arranged in two domains (blue and red). Only main chain (N, C', C $_{\alpha}$ ) atoms and no side chains are shown.



**Figure 5.12** Schematic diagram of the path of the polypeptide chain in one domain (the blue region in Figure 5.11) of the  $\gamma$ -crystallin molecule. The domain structure is built up from two  $\beta$  sheets of four antiparallel  $\beta$  strands, sheet 1 from  $\beta$  strands 1, 2, 4, and 7 and sheet 2 from strands 3, 5, 6, and 8.



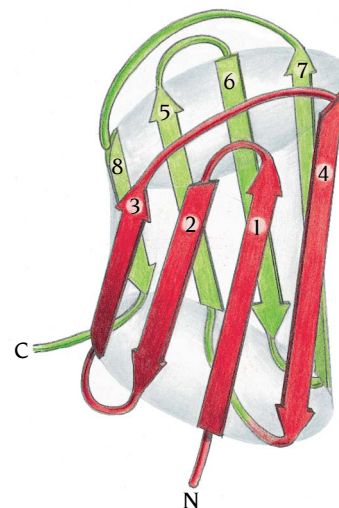
**Figure 5.13** A preliminary topological diagram of the structure of one domain of  $\gamma$  crystallin shown in Figure 5.12, illustrating that the two  $\beta$  sheets are separate within the domain.

look unnecessarily complicated, but notice from the schematic diagram of the domain in Figure 5.12 that the two  $\beta$  sheets are packed against each other so that they form a distorted barrel. To see if the diagram can be simplified, we idealize the barrel and plot the strands along the surface of the barrel as shown in Figure 5.14. It is then immediately obvious that strands 1, 2, 3, and 4 form a Greek key motif, as do strands 5, 6, 7, and 8. These two motifs are joined by a loop across the bottom of the barrel, between strands 4 and 5.

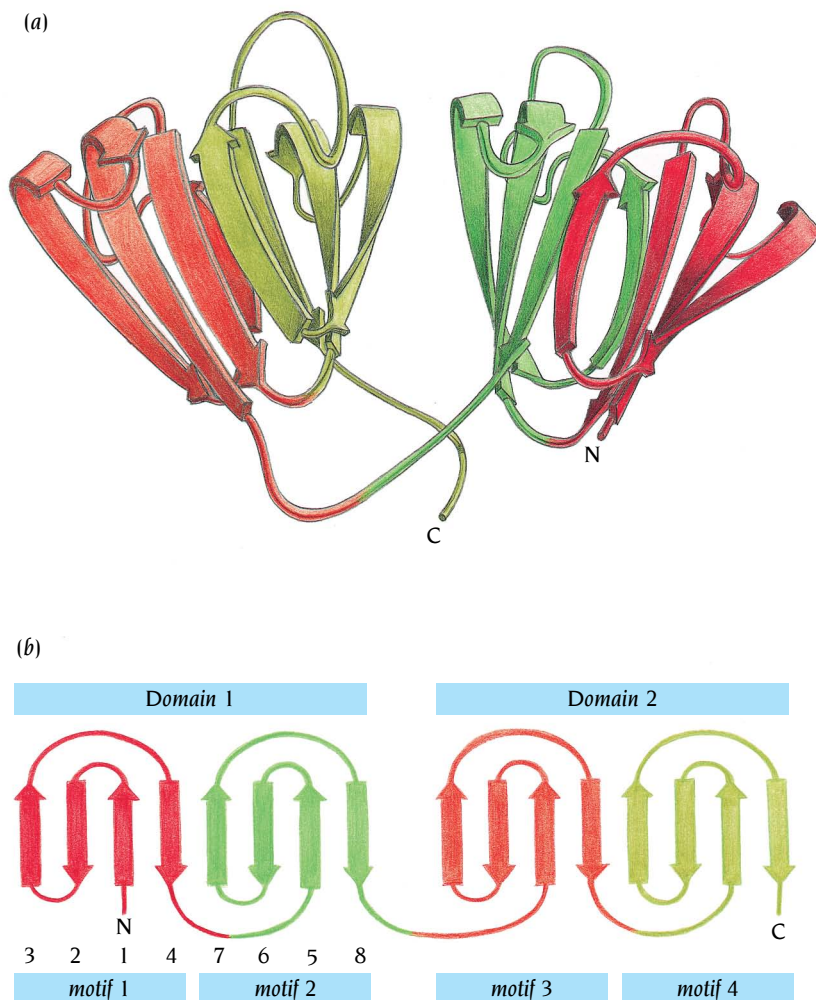
On the basis of this new insight we can draw the topology diagram shown in the left half of Figure 5.15b. What is the difference between this and the previous topological diagram we made? The only changes we have made are to move  $\beta$  strand 3 from the right edge to the left edge of the domain topology and to close the gap between strands 7 and 6. We have changed neither the strand order nor the connections between the strands; thus the two diagrams are topologically identical.

### *The two domains have identical topology*

Using a graphics display, we could do the same thing for the second domain and arrive at the full topology diagram in Figure 5.15b. From this diagram it is obvious that the two domains have identical topology and thus in all probability similar structures. This realization is not at all trivial. To be able



**Figure 5.14** The eight  $\beta$  strands in one domain of the crystallin structure in this idealized diagram are drawn along the surface of a barrel. From this diagram it is obvious that the  $\beta$  strands are arranged in two Greek key motifs, one (red) formed by strands 1–4 and the other (green) by strands 5–8. Notice that the  $\beta$  strands that form one motif contribute to both  $\beta$  sheets as shown in Figure 5.12.



**Figure 5.15** Schematic diagram (a) and topology diagram (b) for the  $\gamma$ -crystallin molecule. The two domains of the complete molecule have the same topology; each is composed of two Greek key motifs that are joined by a short loop region. [(a) Adapted from T. Blundell et al., *Nature* 289: 771–777, 1981.]

to see it when one looks at the structure on the display requires considerable experience because the two domains are in different orientations in the molecule. The brain therefore has to store the image of one domain while examining different orientations of the second domain. A topology diagram, on the other hand, immediately reveals similarities in domain structures. This illustrates one very important use of topology diagrams—namely, to reduce a complicated pattern to a simpler one, from which conclusions can be drawn that are also valid for the complicated pattern.

### *The two domains have similar structures*

A relevant question to ask at this stage is, do the topological identities displayed in the diagram reflect structural similarity? We can now see that topologically the polypeptide chain is divided into four consecutive Greek key motifs arranged in two domains. How similar are the domain structures to each other, and how similar are the two motifs within each domain?

Tom Blundell has answered these questions by superposing the  $C_{\alpha}$  atoms of the two motifs within a domain with each other and by superposing the  $C_{\alpha}$  atoms of the two domains with each other. As a rule of thumb, when two structures superpose with a mean deviation of less than 2 Å they are considered structurally equivalent. For each pair of motifs Blundell found that 40  $C_{\alpha}$  atoms superpose with a mean distance of 1.4 Å. These 40  $C_{\alpha}$  atoms within each motif are therefore structurally equivalent. Since each motif comprises only 43 or 44 amino acid residues in total, these comparisons show that the structures of the complete motifs are very similar. Not only are the individual motifs similar in structure, but they are also pairwise arranged into the two domains in a similar way since superposition of the two domains showed that about 80  $C_{\alpha}$  atoms of each domain were structurally equivalent.

This structural similarity is also reflected in the amino acid sequences of the domains, which show 40% identity. They are thus clearly homologous to each other. The motif structures within the domains superpose equally well but their sequence homology is less, being around 30% between motifs 1 and 2 and 20% between 3 and 4. This study, however, clearly shows that the topological description in terms of four Greek key motifs is also valid at the structural and amino acid sequence levels.

### *The Greek key motifs in $\gamma$ crystallin are evolutionarily related*

These comparisons strongly suggest that the four Greek key motifs are evolutionarily related. We can guess from the amino acid sequence comparison that this protein evolved in two stages, beginning with the duplication of a primordial gene coding for one motif of about 40 amino acid residues, followed by fusion of the duplicated genes to give a single gene encoding one domain. The gene for this domain, we may imagine, later duplicated in turn and fused to give the full gene for the present-day  $\gamma$ -crystallin polypeptide. The evidence that this was the second step lies in the fact that the amino acid sequence homology is greater between the domains than between the motifs within each domain.

There is some circumstantial evidence in the organization of the crystallin gene for the evolutionary history that we have reconstructed. The amino acid sequence of a mouse  $\beta$  crystallin is homologous to that of  $\gamma$  crystallin and shows the same four homologous motifs. Its coding sequence is in separate DNA sequences (exons) interrupted by noncoding DNA sequences (introns). Walter Gilbert at Harvard University suggested in 1978 that genes for large proteins might have evolved by the accidental juxtaposition of exons coding for specific functions. In  $\beta$  crystallin the three introns are positioned at the junctions between the four motifs, supporting Gilbert's ideas. These introns could, therefore, be evolutionary remnants of the gene duplication and fusion events.

### *The Greek key motifs can form jelly roll barrels*

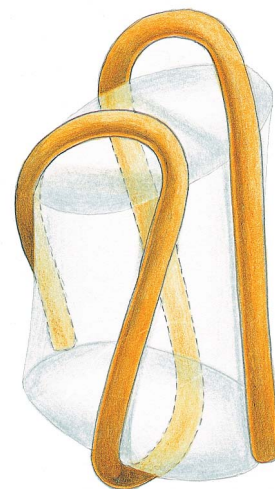
In antiparallel barrel structures with the Greek key motif one of the connections in the motif is made across one end of the barrel. Such connections can be made several times in a  $\beta$  barrel giving different variations and combinations of the Greek key motif. In the structure of crystallin there are two consecutive Greek key motifs that form a barrel with two such connections. There is a different but frequently occurring motif, the **jelly roll motif**, in which there are four connections of this type. It is called jelly roll, or Swiss roll, because the polypeptide chain is wrapped around a barrel core like a jelly roll. This motif has been found in a variety of different structures including the coat proteins of most of the spherical viruses examined thus far by x-ray crystallography, the plant lectin concanavalin A and the hemagglutinin protein from the influenza virus.

### *The jelly roll motif is wrapped around a barrel*

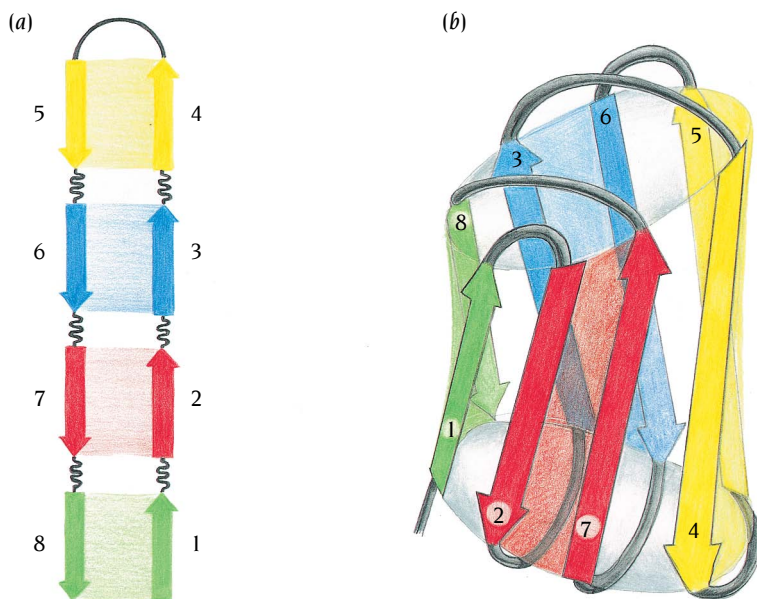
To illustrate how this rather complicated structure is built up, we will start by wrapping a piece of string around a barrel as shown in Figure 5.16. The string goes up and down the barrel four times, crosses over once at the bottom and twice at the top of the barrel. This configuration is the basic pattern for the jelly roll motif.

Let us do the same thing with a strip of paper the width of which is approximately one-eighth of the circumference of the top of the barrel. We imagine that a polypeptide chain follows the edges of this strip, starting at the bottom right corner of the strip and ending at the bottom left corner (Figure 5.17a). The polypeptide chain has eight straight sections,  $\beta$  strands, interrupted by loop regions. The  $\beta$  strands are arranged in a long antiparallel hairpin such that strand 1 is hydrogen bonded to strand 8, strand 2 to strand 7, and so on.

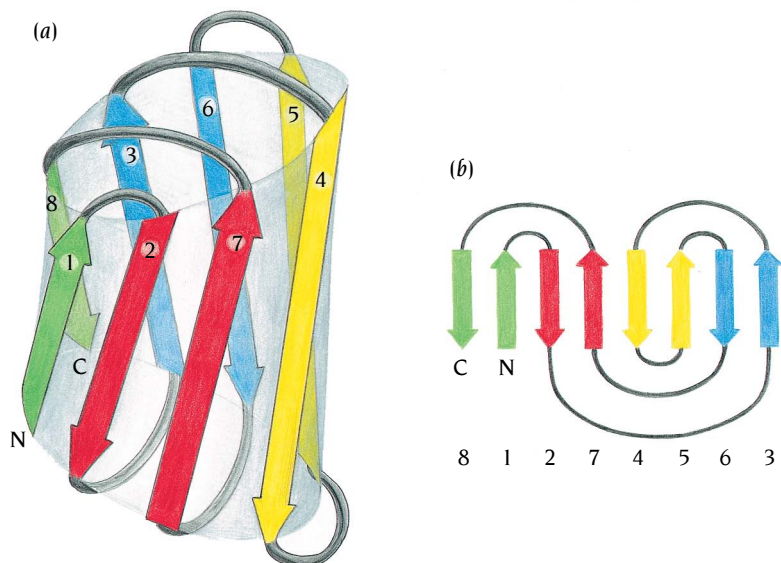
We now wrap the strip around the barrel following the path of the string in Figure 5.16 and in such a way that the  $\beta$  strands go along the sides of the barrel and the loop regions form the connections at the top and bottom of the barrel (Figure 5.17b).



**Figure 5.16** A diagram of a piece of string wrapped around a barrel to illustrate the basic pattern of a jelly roll motif.



**Figure 5.17** A simple illustration of the way eight  $\beta$  strands are arranged in a jelly roll motif. (a) The eight  $\beta$  strands are drawn as arrows along two edges of a strip of paper. The strands are arranged such that strand 1 is opposite strand 8, etc. The  $\beta$  strands are separated by loop regions. (b) The strip of paper in (a) is wrapped around a barrel in the same way as the string in Figure 5.16, such that the  $\beta$  strands follow the surface of the barrel and the loop regions (gray) provide the connections at both ends of the barrel. The  $\beta$  strands are now arranged in a jelly roll motif.



**Figure 5.18** Topological diagrams of the jelly roll structure. The same color scheme is used as in Figure 5.17.

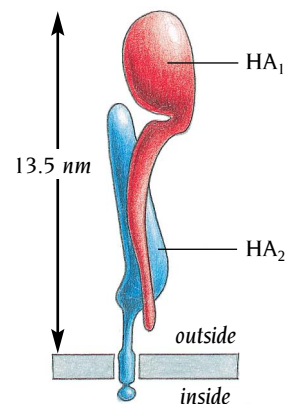
The hydrogen-bonded antiparallel  $\beta$  strand pairs 1:8, 2:7, 3:6, and 4:5 are now arranged such that  $\beta$  strand 1 is adjacent to strand 2, 7 is adjacent to 4, 5 to 6, and 3 to 8. These can also form hydrogen bonds to each other. All adjacent  $\beta$  strands are antiparallel. This is the basic jelly roll  $\beta$ -barrel structure for eight  $\beta$  strands (Figure 5.18a). Most such barrels have eight strands, but any even number of strands greater than four can form a jelly roll barrel. In eight-stranded barrels there are two connections across the top of the barrel and two across the bottom. In addition, there are two connections between adjacent  $\beta$  strands at the top and one at the bottom. A topological diagram of this fold is given in Figure 5.18b.

The jelly roll barrel is thus conceptually simple, but it can be quite puzzling if it is not considered in this way. Discussion of these structures will be exemplified in this chapter by hemagglutinin and in Chapter 16 by viral coat proteins.

### *The jelly roll barrel is usually divided into two sheets*

The barrels we have used to illustrate both the Greek key and the jelly roll structures provide topological descriptions, as defined in Chapter 2. A topological description accurately represents the connectivity and the strand order around the barrel and thus is very useful in the same way that a subway map tells you how stations are interconnected. However, when one analyzes the pattern of hydrogen bonds between the  $\beta$  strands of such barrels, one finds that they usually form two sheets with few if any hydrogen bonds between strands that belong to the different  $\beta$  sheets, as we saw in the crystallin structure. The barrel is distorted and adjacent  $\beta$  strands are separated from each other in two places across the barrel. The division of  $\beta$  strands into these two sheets does not necessarily follow the division into topological motifs. The  $\beta$  strands in jelly roll barrels are also usually arranged in two sheets that are packed against each other. This does not, however, change either the topology or the usefulness of the description of these structures as barrels as long as one keeps in mind that these barrels are distorted and flattened.

**Figure 5.19** Schematic picture of a single subunit of influenza virus hemagglutinin. The two polypeptide chains HA<sub>1</sub> and HA<sub>2</sub> are held together by disulfide bridges.



### *The functional hemagglutinin subunit has two polypeptide chains*

We have already discussed one envelope protein of influenza virus, neuraminidase, as an example of an up-and-down antiparallel  $\beta$  motif. In the second envelope protein, hemagglutinin, one domain of the polypeptide chain is folded into a jelly roll motif. We shall now look at some other features of hemagglutinin that are important for its biological function.

The hemagglutinin polypeptide chain is synthesized on membrane-bound ribosomes of the rough endoplasmic reticulum and then cotranslationally inserted into the membrane. The polypeptide chain is proteolytically cleaved to yield two chains of 328 and 221 amino acids called HA<sub>1</sub> and HA<sub>2</sub>, respectively, which are held together by disulfide bonds (Figure 5.19). Three hemagglutinin monomers, each with one HA<sub>1</sub> and one HA<sub>2</sub> chain, trimerize in the rough endoplasmic reticulum and are transported from there through the Golgi apparatus to the plasma membrane, where the functional part of the molecule is now outside the cell anchored to the cell membrane via the HA<sub>2</sub> tails.

Progeny virus particles then bud from patches of the infected cell's plasma membrane that contain both the viral hemagglutinin and neuraminidase. The viral envelopes therefore contain both viral membrane proteins but no cellular membrane proteins.

Protease treatment of influenza virus particles cleaves the three HA<sub>2</sub> chains of the trimeric hemagglutinin molecules. Such cleavage leaves three C-terminal chains, 47 amino acids long, still inserted into the viral envelope and releases a second much larger soluble fragment. The soluble fragment consists of three complete HA<sub>1</sub> chains disulfide bonded to three HA<sub>2</sub> chains complete except for their membrane anchor regions. This soluble trimeric fragment has been crystallized at neutral pH and its structure was determined at 3 Å resolution and subsequently refined to a high resolution in the laboratory of Don Wiley at Harvard University. The influenza virus that he used was the Hong Kong 1968 strain, which caused the "Asian flu" pandemic disease.

### *The subunit structure is divided into a stem and a tip*

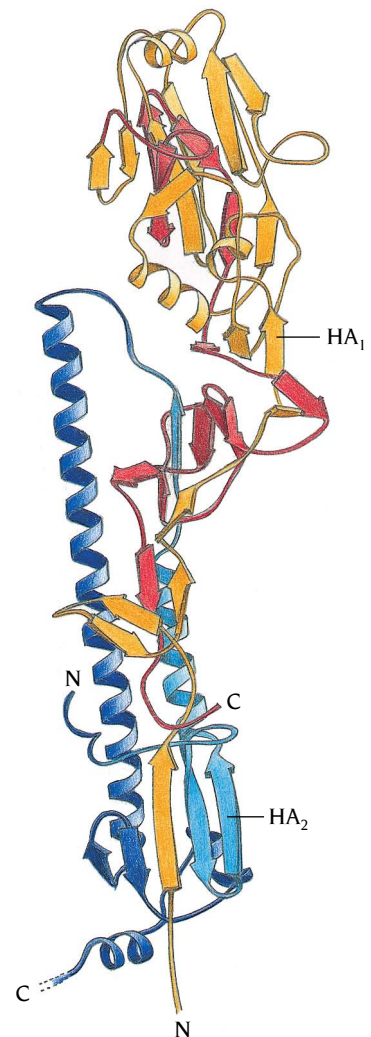
The monomeric subunit is divided into a long, fibrous stemlike region extending outward from the membrane with a globular region at its tip (Figure 5.20). The globular region contains only residues of HA<sub>1</sub>, while the stem contains some residues of HA<sub>1</sub> and all of HA<sub>2</sub>.

The amino terminus of HA<sub>1</sub> is found at the base of the stem close to the viral membrane. The first 63 amino acids of HA<sub>1</sub> (pale red in Figure 5.20) reach, in an almost fully extended structure, nearly 100 Å along the length of the molecule before the first compact fold. These 63 residues form part of the stem region of the subunit. The globular tip is an eight-stranded distorted jelly roll structure comprising residues 116 to 261, which are folded into a distorted barrel. The remaining 70 residues of HA<sub>1</sub> return to the stem region, running nearly antiparallel to the initial stretch of 63 residues.

The major structural feature of the HA<sub>2</sub> chain (blue in Figure 5.20) is a hairpin loop of two  $\alpha$  helices packed together. The second  $\alpha$  helix is 50 amino acids long and reaches back 76 Å toward the membrane. At the bottom of the stem there is a  $\beta$  sheet of five antiparallel strands. The central  $\beta$  strand is from HA<sub>1</sub>, and this is flanked on both sides by hairpin loops from HA<sub>2</sub>. About 20 residues at the amino terminal end of HA<sub>2</sub> are associated with the activity by which the virus penetrates the host cell membrane to initiate infection. This region, which is quite hydrophobic, is called the fusion peptide.

The hemagglutinin trimer molecule is 135 Å long (from membrane to tip) and varies in cross-section between 15 Å and 40 Å. It is thus an unusually

**Figure 5.20** Schematic diagram of the subunit structure of hemagglutinin from influenza virus. The structure comprises about 550 amino acids arranged in two chains HA<sub>1</sub> (red) and HA<sub>2</sub> (blue). The first half of each chain has a lighter color in the diagram. The subunit is very elongated with a long stemlike region built up by residues from both chains and includes one of the longest  $\alpha$  helices known in a globular structure, about 75 Å long. The globular head is formed by residues only from HA<sub>1</sub>. (Courtesy of Don Wiley, Harvard University.)



elongated molecule. The long fibrous stems of each subunit form the major subunit contacts (Figure 5.21). In particular, the three long HA<sub>2</sub> helices, one from each subunit, intertwine by forming a coiled-coil structure (see Chapter 3) and pack against each other for part of their lengths forming a core 40 Å long stabilized by both hydrophobic residues and internal salt bridges. In addition the three heads of the subunits are close together and interact with each other, further stabilizing the trimer.

### The receptor binding site is formed by the jelly roll domain

To initiate infection, the virus hemagglutinin binds to the sialic acid residues of glycosylated receptor proteins on the target cell surface. Once bound to the receptor, the virus is then taken into the cell by endocytosis. The receptor binding site on the hemagglutinin molecule has been determined from experiments with mutants of hemagglutinin and from the determination of structures with bound inhibitors (modified sialic acid molecules with substituents at two different positions of the sugar ring, Figure 5.22).

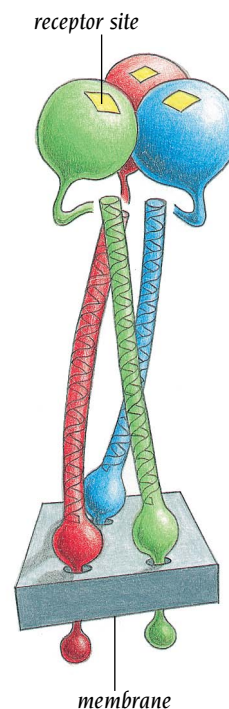
The binding site is located at the tip of the subunit within the jelly roll structure (Figure 5.23). The sialic acid moiety of the hemagglutinin inhibitors binds in the center of a broad pocket on the surface of the barrel (Figure 5.24). In addition to this groove there is a hydrophobic channel that can accommodate large hydrophobic substituents at the C<sub>2</sub> position of sialic acid (Figures 5.22 and 5.24).

Antibodies in our immune system bind to the receptor binding site, so preventing the virus from entering a cell. The virus can escape this neutralization through mutations in residues that form this binding site. Such mutations, however, are found only at the rim of the sialic acid binding pocket, presumably because mutation of residues inside this pocket would prevent the virus from binding to the cell surface receptor and consequently prevent viral propagation. The receptor binding site is therefore an ideal target for drug design, and studies of inhibitor binding to this site have provided valuable clues and ideas for the design of molecules that are candidate drug therapies for influenza virus infections.

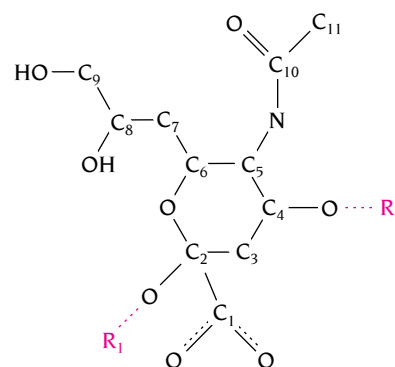
### Hemagglutinin acts as a membrane fusogen

In addition to binding to sialic acid residues of the carbohydrate side chains of cellular proteins that the virus exploits as receptors, hemagglutinin has a second function in the infection of host cells. Viruses, bound to the plasma membrane via their membrane receptors, are taken into the cells by endocytosis. Proton pumps in the membrane of endocytic vesicles that now contain the bound viruses cause an accumulation of protons and a consequent lowering of the pH inside the vesicles. The acidic pH (below pH 6) allows hemagglutinin to fulfill its second role, namely, to act as a membrane fusogen by inducing the fusion of the viral envelope membrane with the membrane of the endosome. This expels the viral RNA into the cytoplasm, where it can begin to replicate.

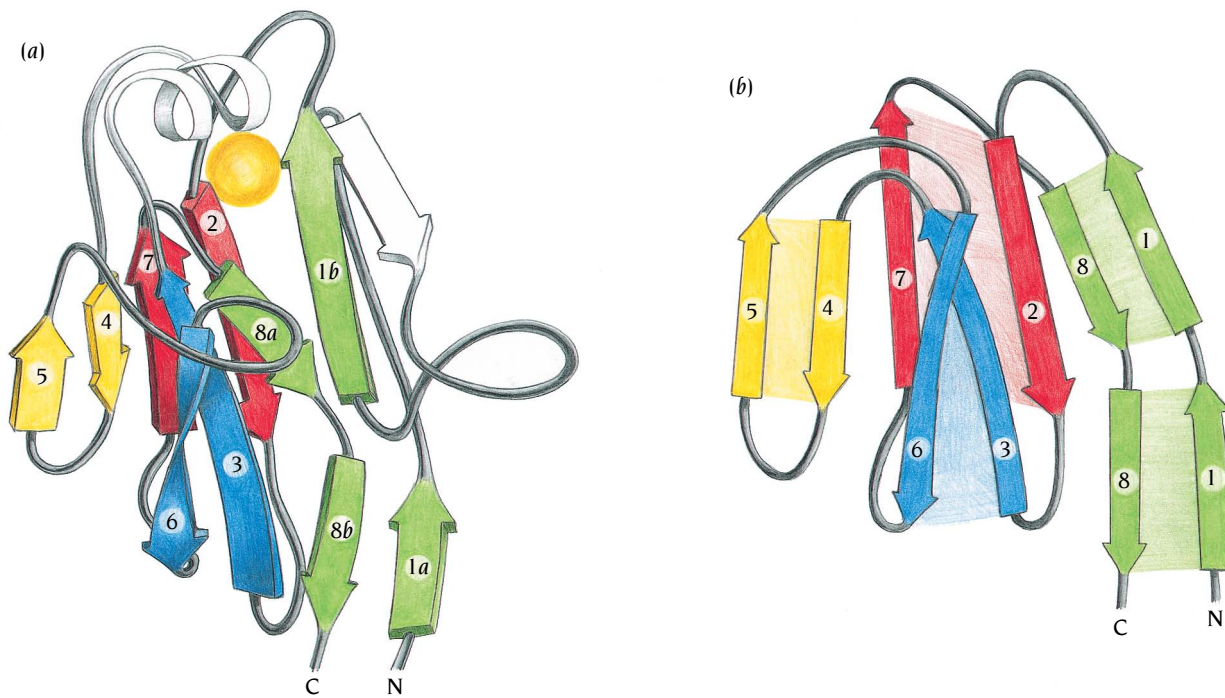
This fusogenic activity of influenza hemagglutinin is frequently exploited in the laboratory. If, for example, the virus is bound to cells at a temperature too low for endocytosis and then the pH of the external medium is lowered, the hemagglutinin causes direct fusion of the viral envelope with the plasma membrane; infection is achieved without endocytosis. Similarly, artificial vesicles with hemagglutinin in their membrane and other molecules in their lumen can be caused to fuse with cells by first allowing the vesicles to bind to the plasma membrane via the hemagglutinin and then lowering the pH of the medium. In this way the contents of the vesicles are delivered to the recipient cell's cytoplasm.



**Figure 5.21** The hemagglutinin molecule is formed from three subunits. Each of these subunits is anchored in the membrane of the influenza virus. The globular heads contain the receptor sites that bind to sialic acid residues on the surface of eukaryotic cells. A major part of the subunit interface is formed by the three long intertwining helices, one from each subunit. (Adapted from I. Wilson et al., *Nature* 289: 366–373, 1981.)



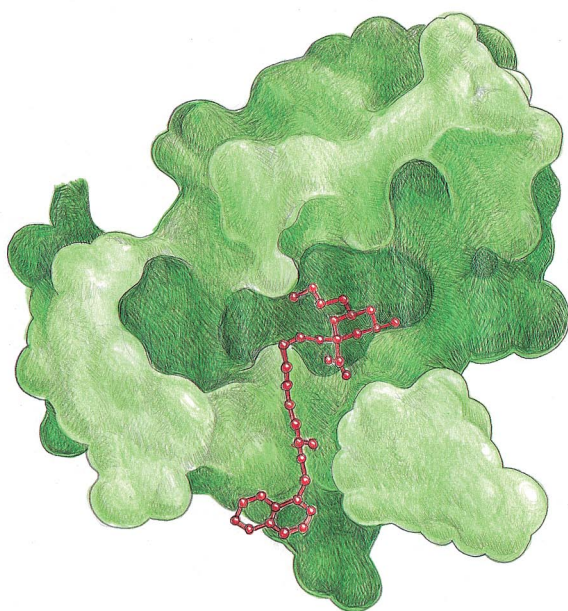
**Figure 5.22** Chemical formula for sialic acid ( $\alpha$ -5-*n*-acetylneuraminic acid) drawn in approximately the same orientation as the ball and stick models in Figure 5.24. R<sub>1</sub> and R<sub>2</sub>, which are H atoms in sialic acid, denote substituents introduced to design tightly bound inhibitors. These are large and hydrophobic as shown in Figure 5.24.



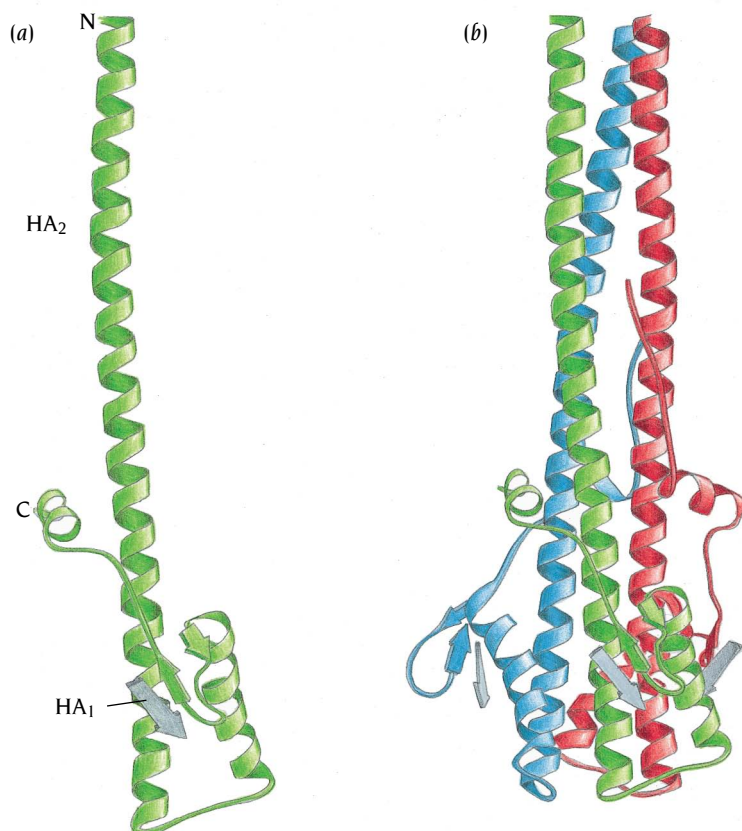
### *The structure of hemagglutinin is affected by pH changes*

The structure of the soluble trimeric hemagglutinin fragment above pH 6 gave no indication of how hemagglutinin induces membrane fusion. When the soluble trimers were exposed to pH below 6 they aggregated and therefore could not be crystallized. However, by using monoclonal antibodies to specific epitopes on the two chains, HA<sub>1</sub> and HA<sub>2</sub>, it was shown that lowering the pH causes a massive conformational change in hemagglutinin. As a result of this induced conformational change the hemagglutinin becomes highly susceptible to proteolytic cleavage. This property was used in the laboratory of Don Wiley to produce a soluble fragment of the low pH form comprising

**Figure 5.23** The globular head of the hemagglutinin subunit is a distorted jelly roll structure (a).  $\beta$  strand 1 contains a long insertion, and  $\beta$  strand 8 contains a bulge in the corresponding position. Each of these two strands is therefore subdivided into shorter  $\beta$  strands. The loop region between  $\beta$  strands 3 and 4 contains a short  $\alpha$  helix, which forms one side of the receptor binding site (yellow circle). A schematic diagram (b) illustrates the organization of the  $\beta$  strands into a jelly roll motif.



**Figure 5.24** Space-filling model (green) of the sialic acid binding domain of hemagglutinin with a bound inhibitor (red) illustrating the different binding grooves. The sialic acid moiety of the inhibitor binds in the central groove. A large hydrophobic substituent, R<sub>1</sub>, at the C<sub>2</sub> position of sialic acid binds in a hydrophobic channel that runs from the central groove to the bottom of the domain. (Adapted from S.J. Watowich et al., *Structure* 2: 719–731, 1994.)



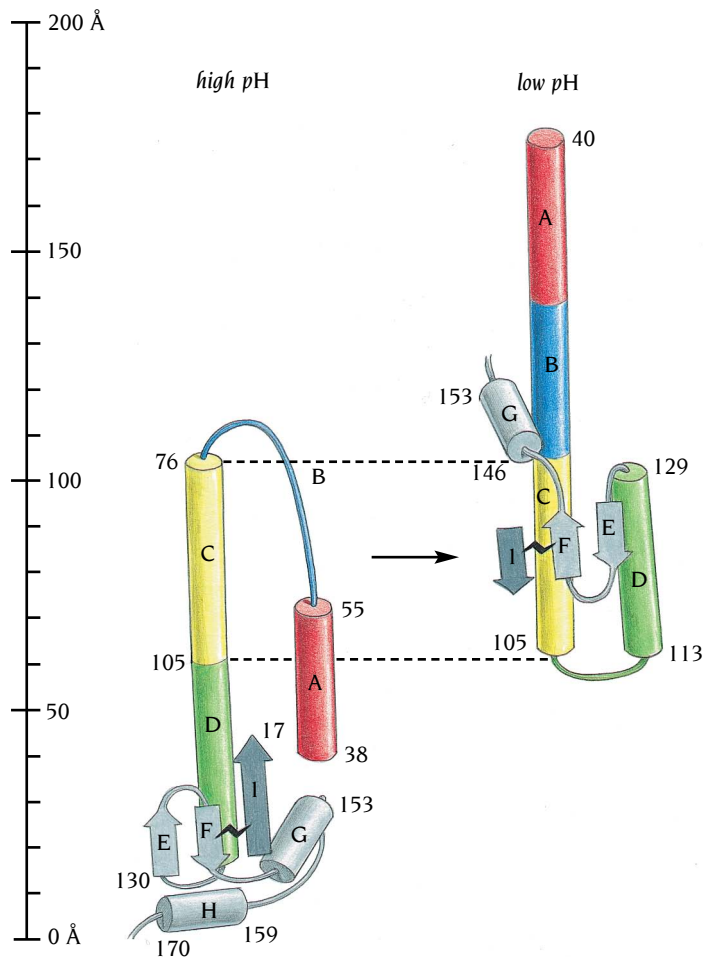
**Figure 5.25** Schematic diagrams of the structure of a proteolytic fragment of hemagglutinin at low pH where the molecule induces membrane fusion. (a) Residues 38–175 of the HA<sub>2</sub> polypeptide chain (green) form a 100-Å long  $\alpha$  helix starting at the N-terminus followed by a loop, a  $\beta$  hairpin and finally a short C-terminal helix. In addition the diagram shows a  $\beta$  strand from the HA<sub>1</sub> polypeptide chain (gray) which participates with the  $\beta$  hairpin to form a three-stranded antiparallel  $\beta$  sheet. (b) The proteolytic fragment forms a trimer like the intact hemagglutinin molecule. The three long  $\alpha$  helices of the three subunits intertwine to form a three-stranded coiled coil. (Adapted from P.A. Bullough et al., *Nature* 371: 37–43, 1994.)

residues 1–27 of the HA<sub>1</sub> chain and residues 38–175 of the HA<sub>2</sub> chain that could be crystallized; its structure was determined in 1994 (Figure 5.25).

The structure of this fragment, which like the high pH fragment is a trimer, confirms that the HA<sub>2</sub> subunit undergoes major structural changes at low pH. Most of the secondary structure elements are essentially preserved but there are two important exceptions (Figure 5.26). First, the loop region B between the two long  $\alpha$  helices A and C + D in the high pH structure changes into an  $\alpha$  helix. Second, an  $\alpha$ -helical region in the middle of helix C + D changes into a loop region (Figure 5.26). The resulting helix A + B + C comprises 65 residues and is about 100 Å long. Previous examination of the amino acid sequence of this whole region, residues 40–106, had shown a very clear heptad repeat typical of coiled-coil structures (see Chapter 3). It was therefore surprising that in the high pH structure only a part of this region, helix C, actually has a coiled-coil structure. Reassuringly, in the low pH structure the whole region is a coiled coil which is involved in trimerization (Figure 5.25).

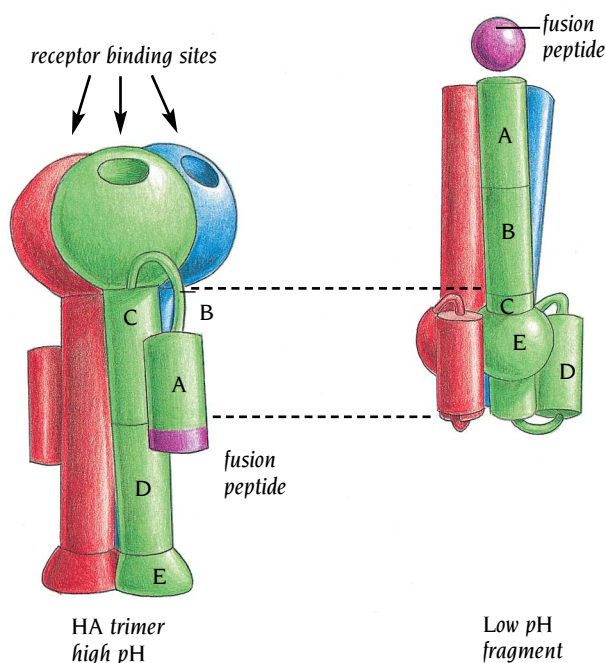
The new loop region between helices C and D changes completely the position and orientation of helix D so that instead of being continuous with helix C it is packed sideways against helix C in an antiparallel fashion (Figure 5.26). The small  $\alpha$  helices and  $\beta$  strands at the C-terminal region of the chain collectively follow the realignment of helix D and occupy totally different positions (Figure 5.26).

The consequences for a possible fusion mechanism of this structural change are significant. In the high pH structure, the fusion peptide is attached to the N-terminus of helix A and is about 100 Å away from the receptor binding site. At low pH the N-terminus of helix A moves about 100 Å and is brought to the same region of the molecule as the receptor binding site (Figure 5.27). Even though the picture is incomplete, since only a proteolytic fragment of the low pH form was studied, it is quite clear that after these structural changes the likely positions of the receptor binding site and the fusion peptide are compatible with a fusion mechanism whereby hemagglutinin brings the viral and cellular membranes close together.



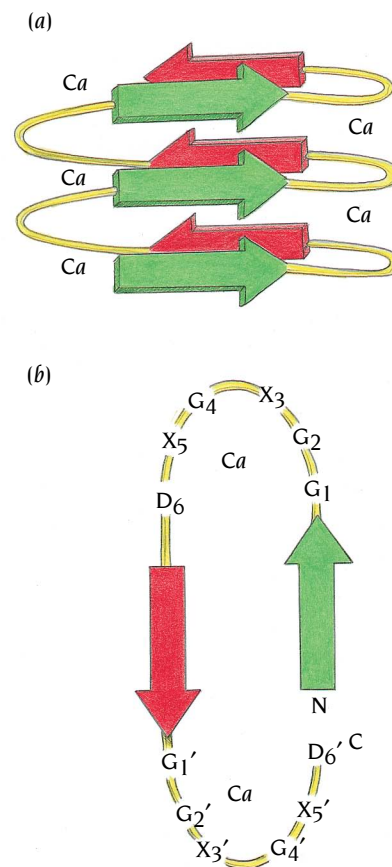
**Figure 5.26** Schematic diagram illustrating the large conformational differences between the high and low pH forms of hemagglutinin. The loop region B (blue) in the high pH form has changed into an  $\alpha$  helix producing a continuous 100-Å-long helix composed of regions A (red), B (blue) and C (yellow) at low pH. Furthermore, residues 105–113, which in the high pH form are in the middle of helix C–D, form a loop in the low pH form causing helix D (green) to be at a very different position. Consequently the  $\beta$ -hairpin E–F and the C-terminal helix G as well as the  $\beta$  strand I from HA<sub>1</sub> occupy very different positions in the two forms even though they have the same internal structure. The squiggle between  $\beta$  strands F and I denotes the S–S bond that joins HA<sub>1</sub> to HA<sub>2</sub>. (Adapted from P.A. Bullough et al., *Nature* 371: 37–43, 1994.)

The large conformational change induced by low pH is irreversible. The low pH form is more thermostable than the high pH form and does not revert to the initial state when pH is raised. This suggests that the low pH form is lower in energy than the high pH form and that the energy required for fusion is stored up during the formation of the hemagglutinin molecule. Each of the three subunits of the trimeric hemagglutinin starts out as a single



**Figure 5.27** Schematic representation of a model for the conformational change of hemagglutinin that at low pH brings the fusion peptide to the same end of the molecule as the receptor binding site. The fusion peptide (purple) is at the end of helix A about 100 Å away from the receptor binding site in the high pH form. In the low pH fragment this region of helix A has moved about 100 Å towards the area where the receptor binding sites are expected to be in the intact hemagglutinin molecule. (Adapted from D. Stuart, *Nature* 371: 19–20, 1994.)

**Figure 5.28** Schematic diagrams of the two-sheet  $\beta$  helix. Three complete coils of the helix are shown in (a). The two parallel  $\beta$  sheets are colored green and red, the loop regions that connect the  $\beta$  strands are yellow. (b) Each structural unit is composed of 18 residues forming a  $\beta$ -loop- $\beta$ -loop structure. Each loop region contains six residues of sequence Gly-Gly-X-Gly-X-Asp where X is any residue. Calcium ions are bound to both loop regions. (Adapted from F. Jurnak et al., *Curr. Opin. Struct. Biol.* 4: 802–806, 1994.)



polypeptide chain, folded into a stable conformation. On cleavage to form the HA<sub>1</sub> and HA<sub>2</sub> chains of the mature molecules the free ends snap 20 Å apart to give the metastable high pH structure. This molecule is like a set trap: lowering the endosomal pH springs the trap, setting in motion a series of events which starts with a large conformational change to bring the fusion peptide into a position where it can engage the target membrane.

### Parallel $\beta$ -helix domains have a novel fold

In the first edition of this book this chapter was entitled “Antiparallel Beta Structures” but we have had to change this because an entirely unexpected structure, the  $\beta$  helix, was discovered in 1993. The  $\beta$  helix, which is not related to the numerous antiparallel  $\beta$  structures discussed so far, was first seen in the bacterial enzyme pectate lyase, the structure of which was determined by the group of Frances Jurnak at the University of California, Riverside. Subsequently several other protein structures have been found to contain  $\beta$  helices, including extracellular bacterial proteinases and the bacteriophage P22 tailspike protein.

In these  $\beta$ -helix structures the polypeptide chain is coiled into a wide helix, formed by  $\beta$  strands separated by loop regions. In the simplest form, the two-sheet  $\beta$  helix, each turn of the helix comprises two  $\beta$  strands and two loop regions (Figure 5.28). This structural unit is repeated three times in extracellular bacterial proteinases to form a right-handed coiled structure which comprises two adjacent three-stranded parallel  $\beta$  sheets with a hydrophobic core in between.

This structural organization has striking similarities to that of  $\alpha/\beta$  proteins, the difference being that the loop- $\alpha$ -helix-loop that connects the parallel  $\beta$  strands in  $\alpha/\beta$  structures is substituted by a loop- $\beta$ -strand-loop in these  $\beta$ -helix structures. Instead of having one  $\beta$  sheet adjacent to a stack of  $\alpha$  helices as in the  $\alpha/\beta$  structures described in the previous chapter,  $\beta$ -helix structures have two parallel  $\beta$  sheets. In  $\alpha/\beta$  structures a twist of about 20° between adjacent  $\beta$  strands is imposed by the packing requirements of the  $\alpha$  helices: in order to pack ridges into grooves, as described in Chapter 3, the  $\alpha$  helices have to be twisted with respect to each other and this forces the  $\beta$  strands also to be twisted. In  $\beta$ -helix structures no such constraint is present and therefore the sheets are almost planar and form straight walls (Figure 5.28a).

The basic structural unit of these two-sheet  $\beta$  helix structures contains 18 amino acids, three in each  $\beta$  strand and six in each loop. A specific amino acid sequence pattern identifies this unit; namely a double repeat of a nine-residue consensus sequence Gly-Gly-X-Gly-X-Asp-X-U-X where X is any amino acid and U is large, hydrophobic and frequently leucine. The first six residues form the loop and the last three form a  $\beta$  strand with the side chain of U involved in the hydrophobic packing of the two  $\beta$  sheets. The loops are stabilized by calcium ions which bind to the Asp residue (Figure 5.28). This sequence pattern can be used to search for possible two-sheet  $\beta$  structures in databases of amino acid sequences of proteins of unknown structure.

A more complex  $\beta$  helix is present in pectate lyase and the bacteriophage P22 tailspike protein. In these  $\beta$  helices each turn of the helix contains three short  $\beta$  strands, each with three to five residues, connected by loop regions. The  $\beta$  helix therefore comprises three parallel  $\beta$  sheets roughly arranged as the three sides of a prism. However, the cross-section of the  $\beta$  helix is not quite triangular because of the arrangement of the  $\beta$  sheets. Two of the sheets are

arranged adjacent to each other as in the two-sheet  $\beta$  helix, and the third  $\beta$  sheet is almost perpendicular to the other two (Figure 5.29). The  $\beta$  strands in these three parallel sheets are connected by three loop regions. One loop (loop *a* in Figure 5.29) is short and always formed by only two residues which have invariant conformations. The other two loops are much longer and vary in size and conformation. These long loops protrude from the  $\beta$  sheets and probably form the active site regions on the external surface of the protein. Since the long loop regions vary in size, the number and type of amino acids in each turn of the  $\beta$ -helix structures vary. Consequently, no specific amino acid sequence pattern has been identified for the three-sheet  $\beta$ -helix structures.

The number of helical turns in these structures is larger than those found so far in two-sheet  $\beta$  helices. The pectate lyase  $\beta$  helix consists of seven complete turns and is 34 Å long and 17–27 Å in diameter (Figure 5.30) while the  $\beta$ -helix part of the bacteriophage P22 tailspike protein has 13 complete turns. Both these proteins have other structural elements in addition to the  $\beta$ -helix moiety. The complete tailspike protein contains three intertwined, identical subunits each with the three-sheet  $\beta$  helix and is about 200 Å long and 60 Å wide. Six of these trimers are attached to each phage at the base of the icosahedral capsid.

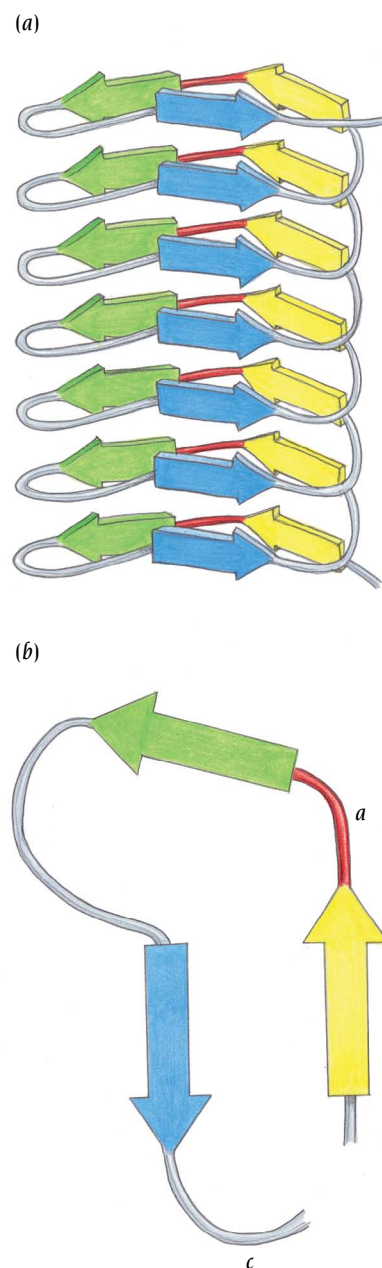
The interior of the  $\beta$ -helix structure in pectate lyase is completely filled with side chains leaving no room for a channel. Interestingly, these side chains are not limited to hydrophobic groups but include polar and charged groups which are all neutralized either by hydrogen bonding or by electrostatic interactions. All the side chains in the interior of the helix are stacked such that the side chains of adjacent turns of the helix have a linear arrangement parallel to the helix axis. These internal stacks fall into different classes; polar stacks of Asn or Ser side chains, aliphatic stacks of Ala, Val, Leu or Ile side chains and aromatic stacks of Phe or Tyr side chains. Pectate lyase is an unusually stable protein and this stacking arrangement no doubt contributes to its stability.

## Conclusion

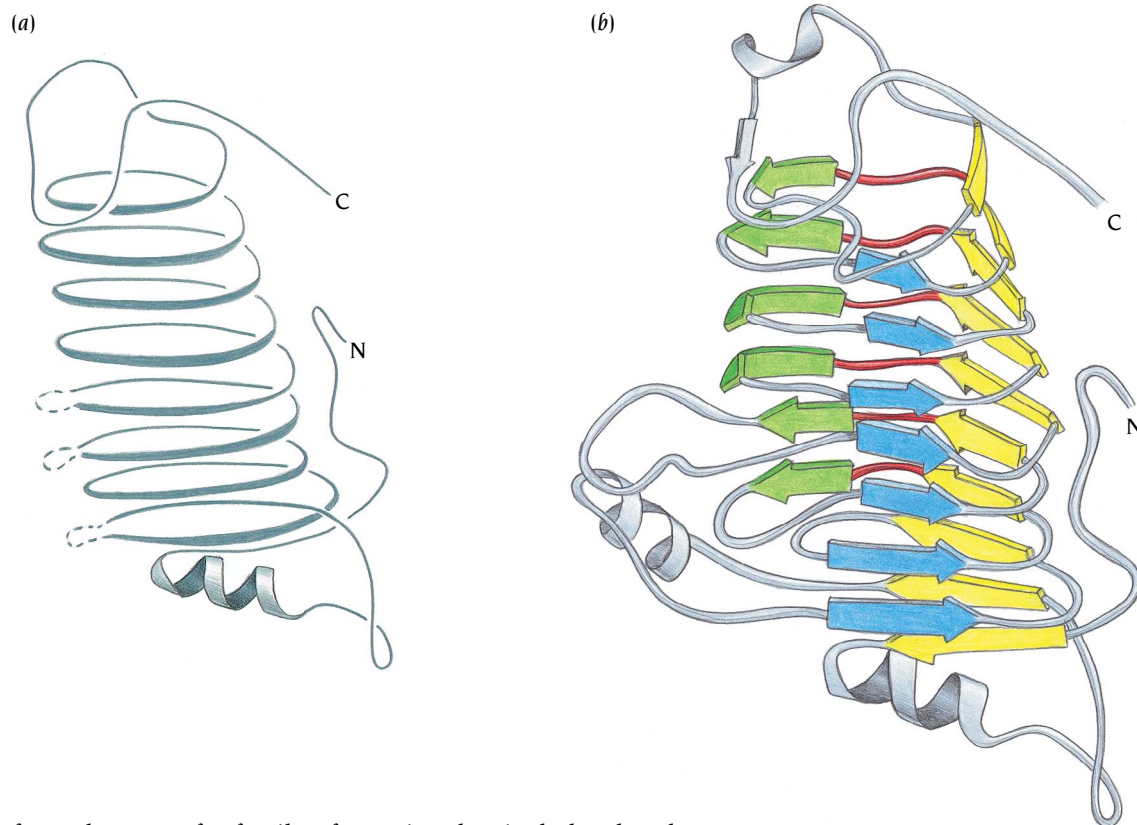
Antiparallel  $\beta$  structures comprise the second major class of protein conformations. In these the antiparallel  $\beta$  strands are usually arranged in two  $\beta$  sheets that pack against each other and form a distorted barrel structure, the core of the molecule. Depending on the way the  $\beta$  strands around the barrel are connected along the polypeptide chain—in other words, depending on the topology of the barrels—they are divided into three main groups: up-and-down barrels, Greek key barrels, and jelly roll barrels.

The number of possible ways to form antiparallel  $\beta$  structures is very large. The number of topologies actually observed is small, and most  $\beta$  structures fall into these three major groups of barrel structures. The last two groups—the Greek key and jelly roll barrels—include proteins of quite diverse function, where functional variability is achieved by differences in the loop regions that connect the  $\beta$  strands that build up the common core region.

Up-and-down barrels are the simplest structures. Each  $\beta$  strand is connected to the next strand by a short loop region. Eight  $\beta$  strands arranged



**Figure 5.29** Schematic diagrams of the three-sheet  $\beta$  helix. (a) The three sheets of parallel  $\beta$  strands are colored green, blue and yellow. Seven complete coils are shown in this diagram but the number of coils varies in different structures. Two of the  $\beta$  sheets (blue and yellow) are parallel to each other and are perpendicular to the third (green). (b) Each structural unit is composed of three  $\beta$  strands connected by three loop regions (labeled *a*, *b* and *c*). Loop *a* (red) is invariably composed of only two residues, whereas the other two loop regions vary in length. (Adapted from F. Jurnak et al., *Curr. Opin. Struct. Biol.* 4: 802–806, 1994.)



**Figure 5.30** Schematic diagrams of the structure of the enzyme pectate lyase C, which has a three-sheet parallel  $\beta$ -helix topology. (a) Idealized diagram highlighting the helical nature of the path of the polypeptide chain which comprises eight helical turns. Dotted regions indicate positions where large external loops have been removed for clarity. (b) Ribbon diagram of the polypeptide chain. The predominant secondary structural elements are three parallel  $\beta$ -sheets which are colored green, blue and yellow. Each  $\beta$ -sheet is composed of 7–10 parallel  $\beta$ -strands with an average length of four to five residues in each strand. The short loop regions of two residues length are shown in red. (Adapted from M.D. Yoder et al., *Science* 260: 1503–1507, 1993.)

in this way form the core of a family of proteins that includes the plasma retinol-binding protein in mammals, biliverdin-binding proteins in insects, and  $\beta$ -lactoglobulin from milk. Members of this family, as well as of the related P2 family with 10  $\beta$  strands, bind large hydrophobic ligands inside the barrel. The barrel seems to be particularly suited to act as a container for chemically quite diverse ligands. Diversity in ligand binding is achieved by differences in the size of the barrel and in the amino acids that also participate in building up the common core.

Most of the known antiparallel  $\beta$  structures, including the immunoglobulins and a number of different enzymes, have barrels that comprise at least one Greek key motif. An example is  $\gamma$  crystallin, which has two consecutive Greek key motifs in each of two barrel domains. These four motifs are homologous in terms of both their three-dimensional structure and amino acid sequence and are thus evolutionarily related.

The jelly roll barrels are found in a variety of protein molecules, including viral coat proteins and hemagglutinin from influenza virus. This structure looks complicated but, in principle, is very simple if one thinks of the analogy of wrapping a strip of paper around a barrel, like a jelly roll. The hemagglutinin receptor-binding domain forms such a jelly roll barrel of eight  $\beta$  strands, where the receptor binding site is at one end of the barrel. During the membrane fusion process of influenza virus infection the hemagglutinin molecule undergoes a major structural change in which the fusion peptide moves about 100 Å to a position close to the receptor binding site.

The second protein in the membrane of influenza virus, neuraminidase, does not belong to any of these three groups of barrel structures. Instead, it forms a propeller-like structure of 24  $\beta$  strands, arranged in six similar motifs that form the six blades of the propeller. Each motif is a  $\beta$  sheet of 4 up-and-down-connected  $\beta$  strands. The enzyme active site is formed by loop regions on one side of the propeller.

In addition to the antiparallel  $\beta$ -structures, there is a novel fold called the  $\beta$  helix. In the  $\beta$ -helix structures the polypeptide chain is folded into a wide helix with two or three  $\beta$  strands for each turn. The  $\beta$  strands align to form either two or three parallel  $\beta$  sheets with a core between the sheets completely filled with side chains.

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